EXAMINATION ENTRY SYSTEM

by

Group IX

Submitted in partial fulfilment of the requirement for the degree of Bachelor of Science in Information Technology

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March, 2025

Declaration

We Group 09, do hereby declare that this Project Report is original and has not been published and/or submitted for any other degree award to any other University before.

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Abstract

The Examination Entry System is a digital solution designed to streamline the exam registration and verification process in universities and faculties. The system enables students to apply for exams seamlessly while allowing management to verify eligibility based on attendance records and disciplinary actions. It ensures transparency and efficiency in handling exam applications by integrating a structured approval workflow.
The system categorizes users into five roles: Admin (Head of the Examination Branch), Dean, Head of Department (HOD), Lecturer in Charge, and Students, each with distinct responsibilities to ensure smooth exam management.
Students can apply for exams through the system, while the Lecturer in Charge verifies their eligibility based on attendance records. The HOD and Dean can oversees this process, ensuring accuracy and fairness and they can update the eligibility based on key concerns, while the Admin (Head of the Examination Branch) maintains overall system control and print the admission and attendance at the end after all the procedures are done.
Additionally, the system enables Admin to generate attendance sheets based on verified exam entries, reducing administrative workload and errors. By digitizing exam registration, the system minimizes paperwork, reduces administrative workload, and ensures accuracy in exam-related processes.This scalable and user-friendly platform enhances efficiency, reduces manual errors, and ensures a smooth exam management experience for students and faculty.

Acknowledgement

We are deeply indebted to our project supervisor Dr. S. Kirushanth whose unlimited steadfast support and inspirations have made this project a great success. In a very special way, we thank him for every support he has rendered unto us to see that we succeed in this challenging study.

Special thanks go to our friends and families who have contained the hectic moments and stress we have been through during the course of the research project.

We thank our University for giving us the grand opportunity to work as a team which has indeed promoted our team work spirit and communication skills. We also thank the individual group members for the good team spirit and solidarity.

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Chapter 1

Introduction

1.1 Background

Managing exam registrations in universities is often a time-consuming process, relying on manual paperwork and multiple approval stages. Inefficiencies in verifying student eligibility based on attendance and disciplinary records can lead to administrative delays and errors. To address these challenges, a digital **Examination Entry System** is proposed. This system streamlines exam applications, eligibility verification, and enhances transparency. By reducing manual workload and improving accuracy, it ensures a more efficient and structured exam management process.

1.2 Problem Statement

Traditional exam registration systems in universities rely on manual processes, leading to inefficiencies, delays, and errors in student eligibility verification. The lack of an integrated system results in increased administrative workload, and difficulty in tracking applications. Without a structured digital solution, ensuring transparency and accuracy in exam entry approval becomes challenging. This project aims to develop an **Examination Entry System** that automates these processes, improving efficiency and reducing administrative burdens.

1.3 Main Objective

- Develop a digital platform to streamline exam registration and verification processes in universities.

- Automate student eligibility verification based on attendance records and disciplinary actions.
- Reduce administrative workload by minimizing manual paperwork and data entry.
- Enhance transparency and accuracy in exam entry approval and record management.
- Implement role-based access control for Admin (DR/SAR/AR), Dean, HOD, Lecturer in Charge, and Students.
- Enable automated attendance sheet generation for eligible exam entries.

1.3.1 Specific Objectives

The specific objectives of the study were:

- To develop a digital solution for streamlining exam registration and verification.
- To carry out a preliminary study on the existing system used in universities for exam entry and verification, identifying inefficiencies and challenges.
- To design the system for automating student exam applications, eligibility verification, and attendance sheet generation.
- To implement the designed system with role-based access for Admin, Dean, HOD, Lecturer in Charge, and Students.
- To carry out the testing process of the implemented system to ensure functionality, accuracy, and efficiency in exam management.

1.4 Scope of the study

This project limited its self to a mechanism for digitized exam registration and verification processes in universities. It focuses on allowing students to apply for exams, enabling lecturers to verify eligibility based on attendance and disciplinary records, and providing Heads of Departments and Deans with oversight of the approval process. The system also includes functionality for generating attendance sheets for verified exam entries. This study does not cover grading, result processing, or other academic assessments beyond exam entry management for the time being.

1.5 Significance of the study

- Enhances efficiency in exam registration and verification processes.
- Reduces manual workload and minimizes errors in administrative tasks.
- Ensures accuracy and transparency in student eligibility verification.
- Provides a structured approval process with role-based access (Admin, Dean, HOD, Lecturer, Students).
- Attendance Sheet printing
- Centralized Final Admission Printing
- Simplifies the generation of attendance sheets for verified exam entries.
- Contributes to digital transformation in university exam management.
- Improves overall coordination and reliability in the exam entry process.

Chapter 2

Literature Review

2.1 Introduction

Digital solutions for academic administration are transforming the efficiency and accessibility of educational institutions by addressing challenges such as manual processing errors, administrative delays, and lack of transparency. By incorporating features such as online application submission, automated eligibility verification, real-time status updates, and centralized data management, these systems can significantly enhance the efficiency of examination registration processes. Studies indicate that automation in administrative workflows can reduce processing times by up to 50 percentage in similar academic environments, enabling institutions to manage high volumes of applications while ensuring compliance with eligibility criteria.[1]

Currently, many universities in Sri Lanka rely on traditional paper-based or semi-digital examination entry systems, which often result in inefficiencies such as delayed approvals, misplaced documents, and a lack of real-time updates for students. While some institutions have adopted partial digitization efforts—such as standalone student portals or limited online forms—these solutions often lack full integration with eligibility verification systems, leading to redundant manual interventions. This highlights the need for a fully integrated Examination Entry System that streamlines administrative workflows and enhances student accessibility.[1]

Globally, successful digital academic management solutions provide valuable insights for Sri Lanka's higher education sector. For instance, India's National Academic Depository (NAD) [2] offers a secure online repository for academic credentials, ensuring seamless verification and reducing administrative overhead.

Similarly, Singapore's Integrated Examination Management System (IEMS)

[3] utilizes real-time data validation and AI-driven fraud detection mechanisms to enhance examination security and efficiency.

These case studies suggest that Examination Entry System could adopt strategies such as automated data validation, AI-based eligibility checks, and blockchain-backed credential verification to ensure transparency and security. [4]

However, several challenges must be addressed for successful implementation. One major obstacle is inconsistent digital literacy among students and staff, which can hinder adoption and usability. Additionally, inadequate IT infrastructure in certain universities, such as limited server capacity and outdated software, poses a challenge to full-scale digitization. Furthermore, concerns regarding data security and privacy require robust encryption and access control mechanisms to protect sensitive student information. Addressing these challenges is critical to ensuring the long-term sustainability of the Examination Entry System. [4]

In conclusion, the Examination Entry System presents an opportunity to modernize academic administration in Sri Lanka's higher education institutions. By leveraging global best practices and adapting them to local challenges, Sri Lanka can enhance the efficiency, transparency, and accessibility of examination registration processes. A well-coordinated strategy that prioritizes technological innovation, user engagement, and data security will be vital in realizing this vision.

Chapter 3

Methodology

3.1 Introduction

This chapter outlines the structured approach taken for the development of the Examination Entry System, ensuring efficiency, accuracy, and transparency in examination registration processes. The Agile methodology was chosen to facilitate iterative development, continuous feedback, and adaptability to institutional needs. Agile's flexibility ensures that system functionalities evolve in response to user requirements, regulatory updates, and technological advancements. Below, we detail the core practices, tools, and workflows followed in this methodology.

3.2 Rationale for Agile Methodology

The decision to adopt Agile over traditional Waterfall or RAD (Rapid Application Development) frameworks was based on:

- Stakeholder Collaboration: Regular interaction with students, faculty members, and administrative staff ensured alignment with actual user needs.

- Iterative Refinement: Modular development allowed continuous testing and integration of critical features (e.g., eligibility verification, application tracking) while accommodating last-minute changes.

- Risk Mitigation: Early identification of potential issues (e.g., data security, system scalability) through ongoing testing and feedback loops.

3.3 Agile Practices and Implementation

Weekly sprints were utilized, with each sprint targeting specific system functionalities:

- Sprint 1-2: Core database design (MySQL) and Next.js frontend prototyping.
- Sprint 3-4: Implementation of student application forms and automated eligibility checks.
- Sprint 5-6: Administrative dashboard, verification workflows, and role-based access controls.
- Sprint 7-8: System testing, security enhancements, and deployment readiness.

3.4 Tools and Collaboration

- Development: VS Code, Next.js(frontend), Node.js(backend), MySQL(backend).
- Version Control: GitHub with branch-per-feature strategy.
- Communication: WhatsApp for developer discussions and Figma for UI collaboration.

3.5 Testing Approach

To ensure the Examination Entry System functions correctly and meets user requirements, both Functional Testing and Ad Hoc Testing were conducted. These testing approaches helped validate system performance and identify any potential issues before deployment.

1. **Functional Testing:** Functional Testing was performed to verify that the system meets its specified requirements. Various input scenarios were designed, and the system's behavior was observed to ensure correctness. Key functionalities tested include:
   - Student Exam Application: Verifying that students can successfully apply for exams.
   - Eligibility Verification: Ensuring that only eligible students are approved based on attendance and disciplinary records.

- Management Dashboard: Testing if staff can view, verify, and process applications without errors.
- Admin Controls: Confirming that the Faculty Dean can oversee and update records as per defined permissions.
- PDF Generation: Checking if verified exam entries can be correctly exported for printing.

Each test case involved providing valid and invalid inputs to confirm that the system correctly processed data and displayed appropriate error messages when necessary.

2. **Ad Hoc Testing:** In addition to structured testing, Ad Hoc Testing was conducted to identify unexpected issues by randomly exploring the system. This type of testing helped uncover potential edge cases and usability concerns that might not have been covered in predefined test cases.

   - Random user interactions were performed to detect UI/UX inconsistencies.
   - Stress scenarios, such as submitting multiple applications at once, were simulated to check system stability.

   This combination of Functional and Ad Hoc Testing ensured that the Examination Entry System operates as expected while also addressing unforeseen issues through exploratory testing.

3.6 Challenges and Adaptations

- User Resistance to Digital Systems: Addressed through training sessions and tutorial videos.
- Security Concerns: Mitigated by implementing role-based access control and data encryption.

3.7 Conclusion

The Agile methodology played a crucial role in balancing technical development with user-centric enhancements. Through iterative sprints, continuous testing, and structured feedback loops, the Examination Entry System achieved its goal of modernizing academic administration while ensuring security and ease of use. This structured approach mitigated risks such as scope creep and system inefficiencies, ultimately contributing to a streamlined and transparent examination registration process.

Chapter 4

System Study, Analysis and Design

4.1 Overview of the System

The Examination Entry System (EES) is a digital platform designed to automate and streamline university exam registration and eligibility verification. It replaces traditional manual processes, reducing errors, improving efficiency, and ensuring transparency in exam management.

4.2 System Study

4.2.1 Existing System Overview

The current exam registration system relies heavily on manual processes, leading to inefficiencies such as:

- Paper-based applications: Students fill out forms that staff process manually.
- Eligibility verification: Staff check attendance and disciplinary records by hand, causing delays.
- Attendance sheet generation: Manual preparation, increasing the risk of errors.
- Limited transparency: Students and faculty have no real-time access to application status.

4.2.2 Existing System Issues

- Manual exam registration leads to delays and errors.
- No centralized verification, making eligibility checks inefficient.
- Staff must manually track attendance and discipline records.
- Students face difficulties tracking their exam application status.
- The administrative workload is high due to paper-based record handling.

4.2.3 Need for the New System

- Reduce manual workload by automating student exam applications.
- Ensure accuracy and transparency in eligibility verification.
- Allow real-time tracking of application status for students and administrators.
- Enable secure and role-based access control for different users.
- Generate automated attendance sheets for exam supervision.

4.2.4 Feasibility Study

1. **Technical Feasibility:**Determines whether the required technology and resources are available to develop the system.
   - Technology Availability: The system use Next.js (React framework) for the Frontend, Node.js for the backend, and MySQL as the database, all of which are widely used and well-supported.
   - Infrastructure: The university already has internet access and computers for students and staff to use the system.
   - Integration Capability: The system can be integrated with existing university databases for student records.
2. **Operational Feasibility:**Determines whether the system meets the needs of users and can be successfully adopted.
   - Ease of Use: The system have a user-friendly dashboard for students and staff to ensure smooth adoption.

- Training Requirements: Minimal training needed as users will interact with simple forms and dropdowns.
- User Acceptance: Students and faculty will likely prefer an automated system over manual processing due to faster exam registration and fewer errors.

3. **Time Feasibility:** The Examination Entry System project was structured into fifteen phases to ensure a systematic and efficient development process:

   - Week 1-2: Project planning, requirement gathering, and finalization of the tech stack.
   - Week 3-4: System architecture design, database schema creation, and UI wireframing.
   - Week 5-8: Parallel development of backend APIs (Node.js, MySQL) and frontend interfaces (Next.js, CSS).
   - Week 9-11: Integration of authentication, role-based access control, and eligibility verification mechanisms.
   - Week 12-13: Functional testing, adhoc testing, and debugging to ensure system reliability.
   - Week 14: User testing and feedback collection from students and administrative staff.
   - Week 15: Final deployment, monitoring, and issue resolution.

4.3 System Analysis

4.3.1 Target Users

- Students
- Admin(DR)
- Management (Dean, Dept.Head, Staff)

4.3.2 Functional Requirements

01. Student Portal:

- Register and apply for exams.

- View eligibility status.

02. Lecture In Charge Portal:

- Verify eligibility based on attendance and disciplinary records.
- Download attendance sheet(temporarily disabled)

03. Department Head Portal:

- Update eligibility if needed
- View Final report sent by Admin

04. Dean Portal:

- Update eligibility if needed
- View Final report sent by Admin

05. Admin Portal:

- Create courses
- Create users
- Create batches and curriculum
- Insert Medical Students and Resit entries
- Verify/Update eligibility if needed
- Generate Examination Admissions
- Generate Attendance Sheets
- Generate Index number for those who don't have Index number

4.3.3 Non-Functional Requirements

- Security: Role-based access control and data encryption.
- Usability: User-friendly interfaces for students and staff.
- Reliability: High uptime and accurate data handling.
- Performance: Fast processing of student applications and verification.

4.3.4 System Workflow(User Role Responsibilities)

- Students log in and apply for exams.
- The staff reviews the applications and confirms eligibility.
- The department heads oversee the verification process and approve the entries for their departments.
- The Dean of Faculty manages the approvals, but cannot delete records.
- DR or Examination Department has full system control but limited deletion permissions.
- The final admissions are generated, and the attendance sheets can be downloaded.

4.3.5 Key Features and Benefits

- Automated Exam Registration: Students can apply for exams online through a secure login.
- Eligibility Verification :- Staff verifies student eligibility based on attendance and discipline records.
- Role-Based Access Control :- Different levels of access for students, staff, department heads, and administrators.
- Attendance Sheet Generation :- Automated creation of exam attendance sheets for staff use.
- Data Security and auditability :- Secure data management and tracking for transparency.
- Centralized Administration :- Efficient handling of exam records and final admission processing.

4.3.6 Use Case Diagram

The use case diagram as shown in the Figure 4.1

4.3.7 ER Diagram

The ER diagram as shown in the Figure 4.2

Figure 4.1: Use Case Diagram

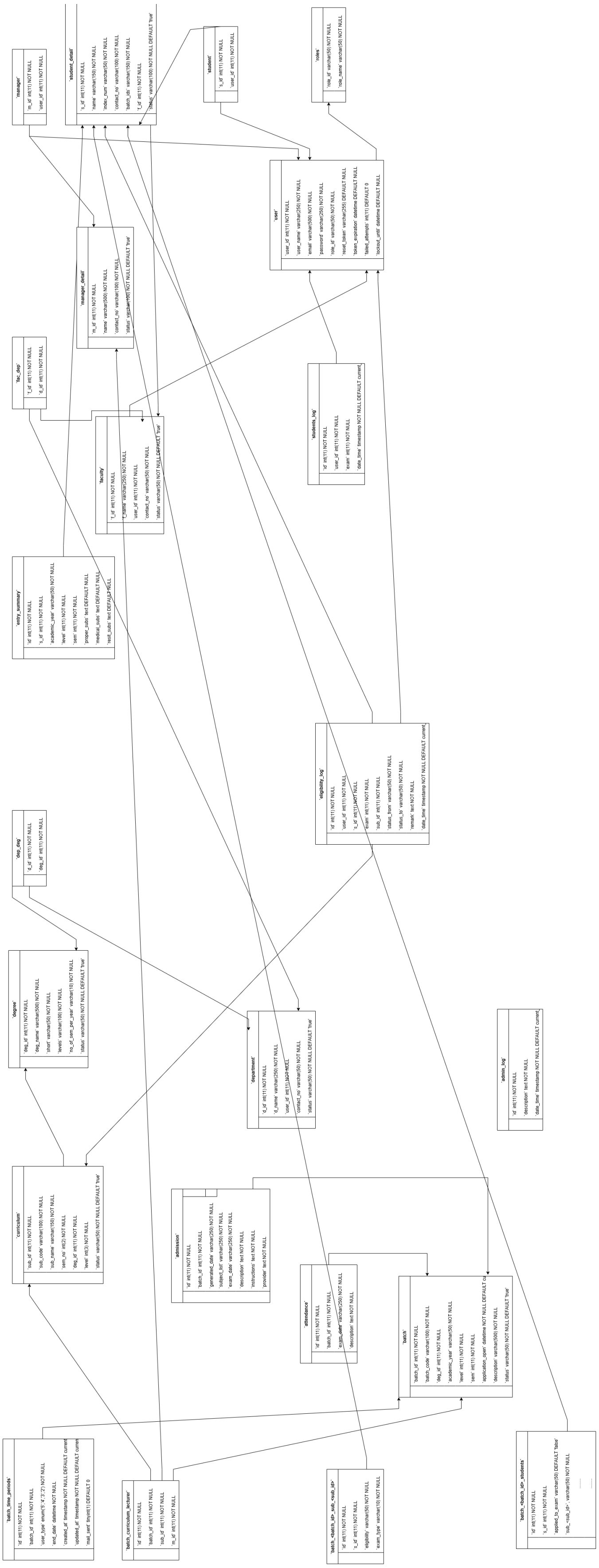

Figure 4.2: ER Diagram

4.4 System Design

4.4.1 Architecture Overview

The system follows a three-tier architecture:

- Frontend: Next.js (React-based framework) for dynamic UI.
- Backend: Node.js to handle requests and business logic.
- Database: MySQL for structured exam-related data storage.

Chapter 5

Presentation of Results

5.1 Introduction

The project cycle as shown in the Figure 5.1

1. Project Initiation (Planning Phase)

- Identify Problems and Objectives:- Analyze the existing manual exam registration process and identify inefficiencies.
- Define Project Scope:- Determine the features, users, and system requirements.
- Feasibility Study:- Assess technical, financial, and operational feasibility.
- Risk Analysis:- Identify potential risks (e.g., data security, system downtime) and mitigation strategies.

2. Requirement Gathering and Documentation

- Functional Requirements:- Define what the system should do (e.g., exam registration, eligibility verification, attendance tracking).
- Non-Functional Requirements:- Set performance, security, and scalability goals.
- User Role Definition:- Identify users (students, staff, department heads, admins) and their responsibilities.
- Data Flow Analysis: Understand how data moves within the system.

3. System Design

- Architecture Design:- Choose a three-tier architecture (Frontend, Backend, Database).
- Database Design:- Create an ER diagram to model entities and relationships
- User Interface (UI/UX) Design:- Develop wireframes and mockups for student and staff dashboards.
- Security and Access Control:- Plan for role-based access control (RBAC) and data protection measures.

4. Implementation and Coding

- Frontend Development: Build the user interface using Next.js, Tailwind CSS, and Shadcn UI.
- Backend Development: Implement business logic using Node.js and MySQL for data storage.
- Integration: Connect frontend, backend, and database for real-time data processing.
- Security Implementation: Encrypt sensitive data and implement authentication mechanisms.

5. Testing

- Unit Functional Testing:- Test individual modules (e.g., student login, exam application, eligibility checks).
- Integration Testing:- Ensure smooth communication between the frontend, backend, and database.
- User Acceptance Testing (UAT):- Get feedback from students and staff to refine the system.
- Performance and Security Testing:- Identify vulnerabilities and optimize for speed and reliability.

6. Deployment

- Pilot Deployment:- Release a beta version to a small group of users for final testing.
- Full Deployment:- Launch the system for all students and staff.
- Training and Documentation: Provide training materials and support to help users adapt.

7. Maintenance

- Bug Fixes and Updates:- Regularly update the system to fix issues and enhance security.
- Performance Monitoring:- Analyze system performance and user feedback for improvements.
- Feature Enhancements:- Implement new features based on university needs.

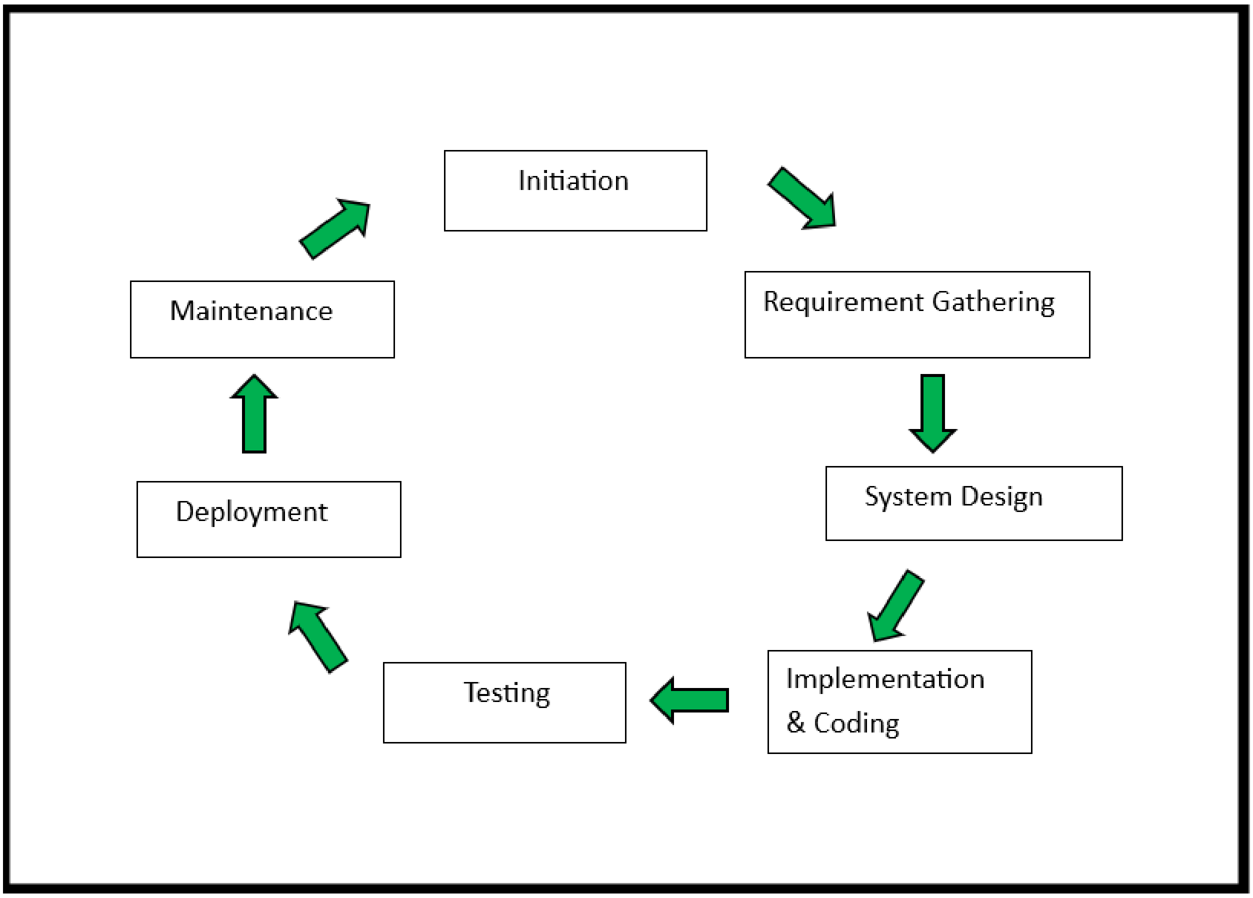

Figure 5.1: Project Cycle

5.2 Screenshots of the system interfaces

- Admin Module

5.2.1 Sign in page

The sign in as shown in the Figure 5.2. The admin can enter the correct username and password,(5.4) then click the "Sign In" button to log in. The "Forgot Password" option allows the admin to reset the password using a reset link. (5.3)

5.2.2 Home page

The Home page as shown in the Figure 5.5. After signing in, the admin can access the Home Page, which includes options such as Batches, Curriculums, Managers, Students, Faculties, Departments, and Degrees.

The right side bar shown in the Figure 5.6. The right side of the sidebar displays the Admin Profile, along with options to Change Password and Log Out.

The Change Password shown in the Figure 5.7. By clicking the "Change Password" option, the admin can update their password by entering the current password, new password, and confirming the new password.

The logout option shown in the Figure 5.8. The Log Out option allows the admin to log out of the system.

Redirects to the home page shown in the Figure 5.9. After logging out, the system redirects to the Home Page.

5.2.3 Dashboard page

The dashboard shown in the Figure 5.10. The Dashboard includes several sections: Home, Courses, Users, Examinations, and Entry Forms.

5.2.4 Courses page

The courses shown in the Figure 5.11. When clicking on Courses, the admin can access buttons for Faculties, Departments, and Degree Programs.

5.12, 5.13, 5.14 shows clicking on Faculty allows the admin to create a new faculty using the "Create Faculty" button.

The admin can modify faculty settings, including setting the status to Active or Not Active. 5.15, 5.16 shown that option.

The Edit option shown in the Figure 5.17. The admin can edit faculty details using the Edit option.

The Search bar shown in the Figure 5.20. The Search Bar enables the admin to search for faculties.

This shown courses in the Figure 5.21. The same actions performed for Faculties can also be done for Departments and Degree Programs.

5.2.5 Users page

Under the Users section of the dashboard, the admin can see buttons for Managers and Students. These sections include similar options as faculties. 5.27, 5.28, 5.29, 5.30, 5.31, 5.32, 5.33 shown that process.

The Import Student shown in the Figure 5.34. Students have an additional option: "Import Student".

The Importing process shown in the Figure 5.35. By selecting the "Import Student" option, the admin can import student details via a CSV file.

The Examinations shown in the Figure 5.36. The next section of the dashboard is "Examinations".

5.2.6 Examinations page

The Examination page shown in the Figure 5.37. This section includes Curriculums and Batches buttons.

The Create subject shown in the Figures 5.38, 5.39, 5.40, 5.41, 5.42, 5.43, 5.44. In Curriculums and Batches, the admin can "Create Subject" and "Create Batch".

The Update batch shown in the Figure 5.45. The admin can update batch details if needed.

The Feed Students shown in the Figure 5.46, 5.47. The "Feed Students" option allows the admin to select students from a list and add them to a batch.

The Update Attendance shown in the Figures 5.48, 5.49 . The "Update Attendance" option enables the admin to upload student attendance using a CSV file.

The Finish Exam shown in the Figure 5.50. After completing the setup, the admin can click the "Finish Exam" button.

The Entry Forms shown in the Figure 5.52. The next dashboard section is "Entry Forms".

5.2.7 Entry Forms page

The Entry Forms page shown in the Figure 5.53. This section includes buttons for Applied Science, Business Studies, and Technological Studies.

The Ready to print shown in the Figure 5.54. A "Ready to Print" option is available.

The "Insert Medical/Resit" option shown in the Figures 5.55, 5.56, 5.57, 5.58, 5.59. If a student has medical or resit subjects, they can apply for these using the "Insert Medical/Resit" option.

The Figure shown Add subjects for Medical/Resit is 5.60.

The Eligibility for exams shown in the Figures 5.61, 5.62, The admin can change a student's eligibility for exams.

The Generate Attendance shown in the Figure 5.63. The "Generate Attendance" option allows the admin to generate an attendance sheet.

The Generate option shown in the Figure 5.65. After filling in the details, the admin can click "Generate" (5.66).

The Generate Admission shown in the Figure 5.67. Finally, by clicking "Generate Admission", the admin can obtain the admission sheet.

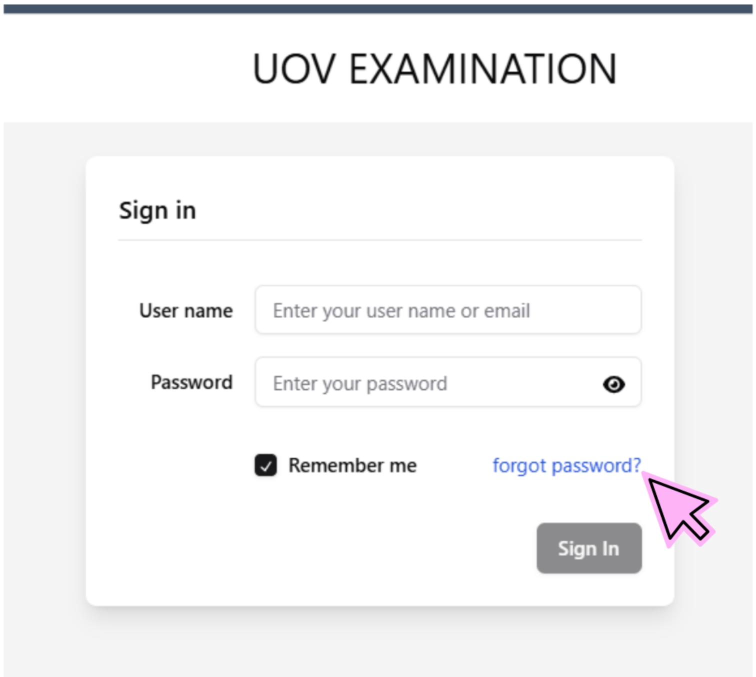

Figure 5.2: Sign in page

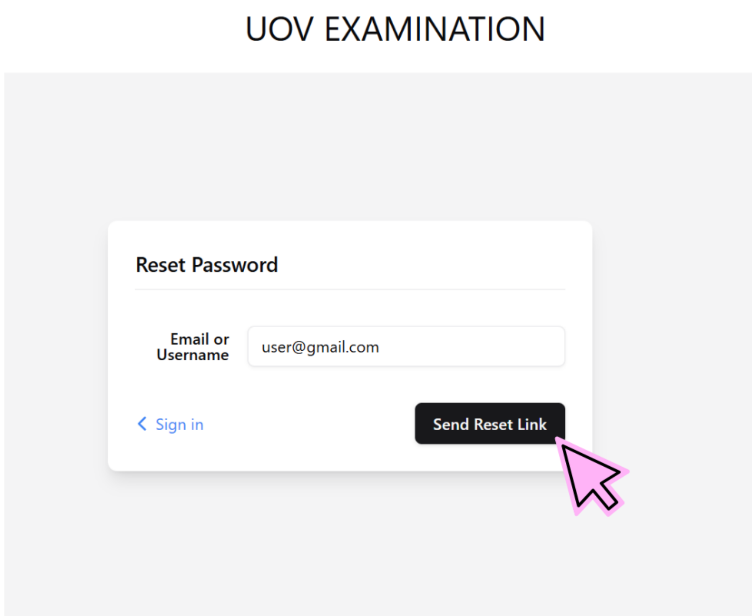

Figure 5.3: Reset Link

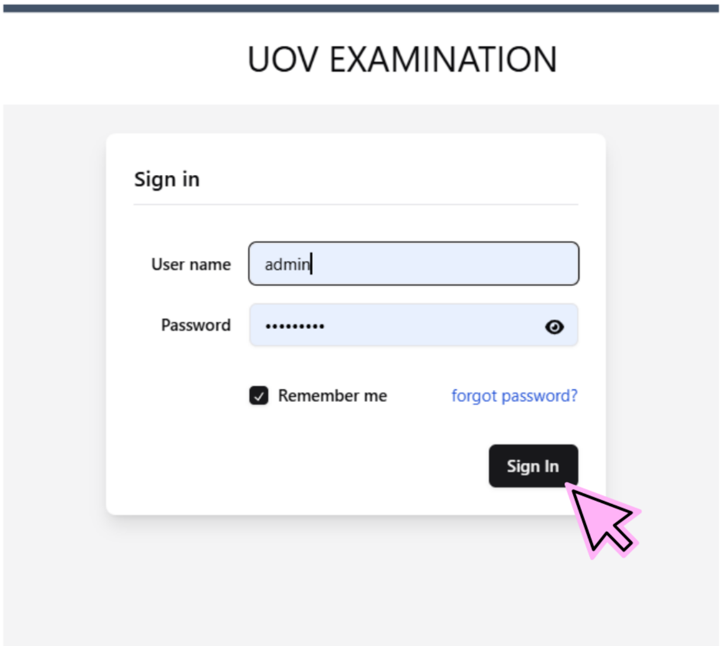

Figure 5.4: Sign in

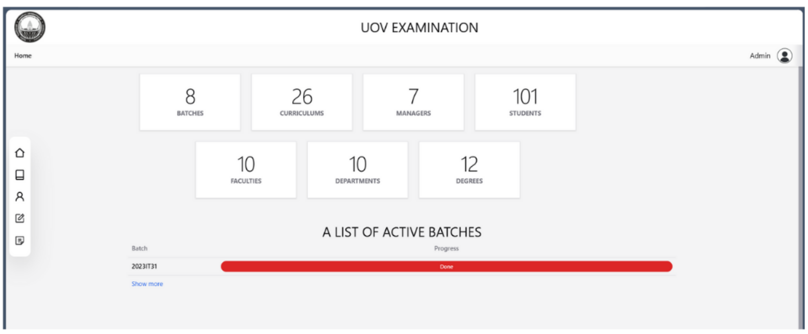

Figure 5.5: Home page

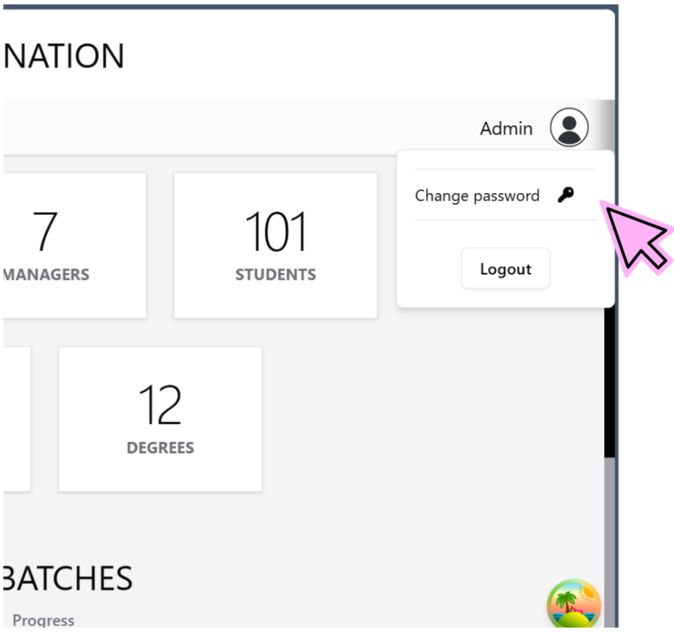

Figure 5.6: Right side bar

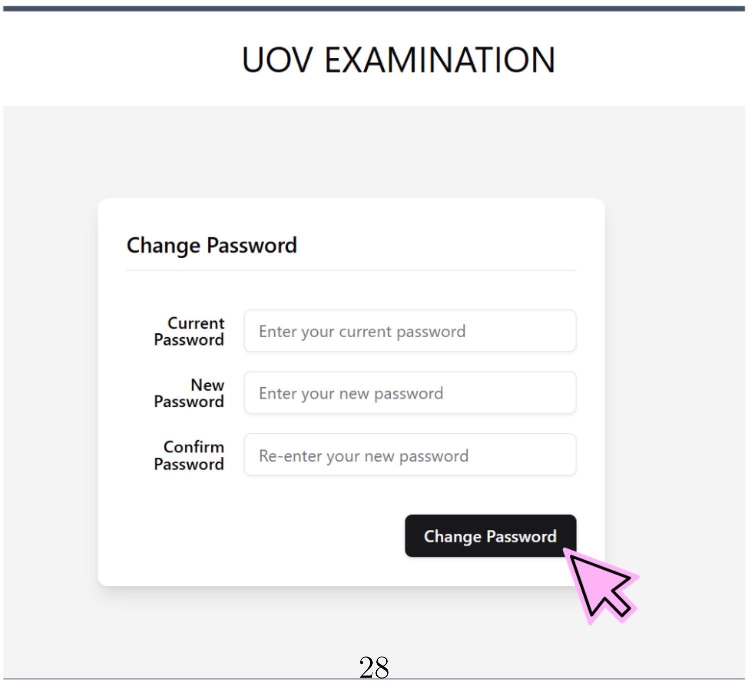

Figure 5.7: Change Password

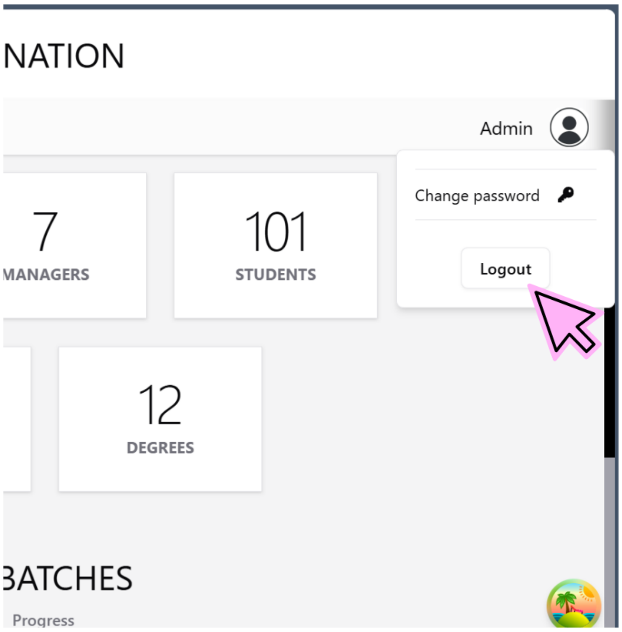

Figure 5.8: Logout option

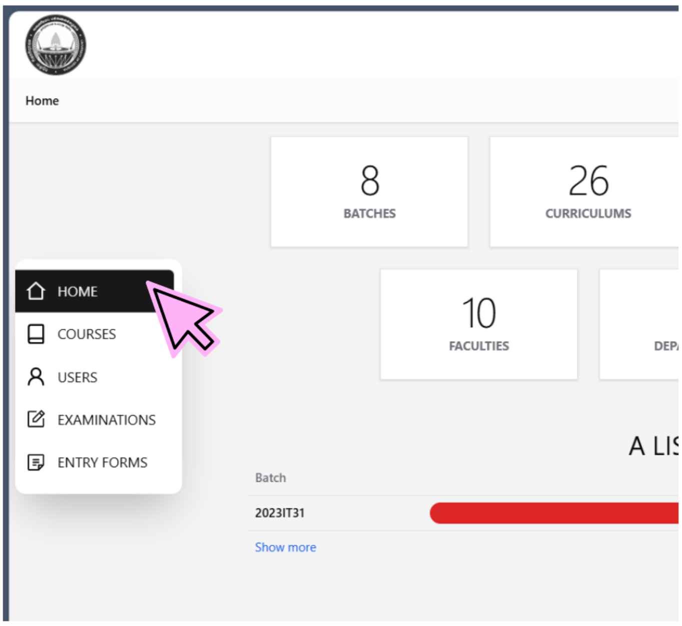

Figure 5.9: Home page

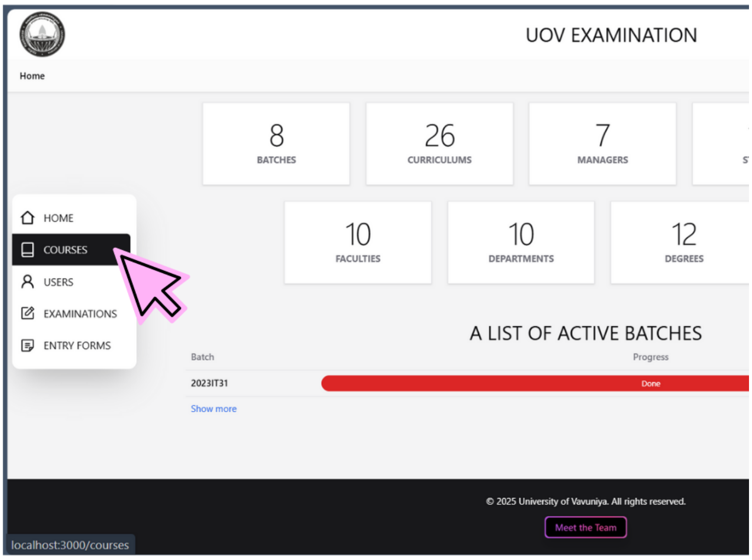

Figure 5.10: Dashboard

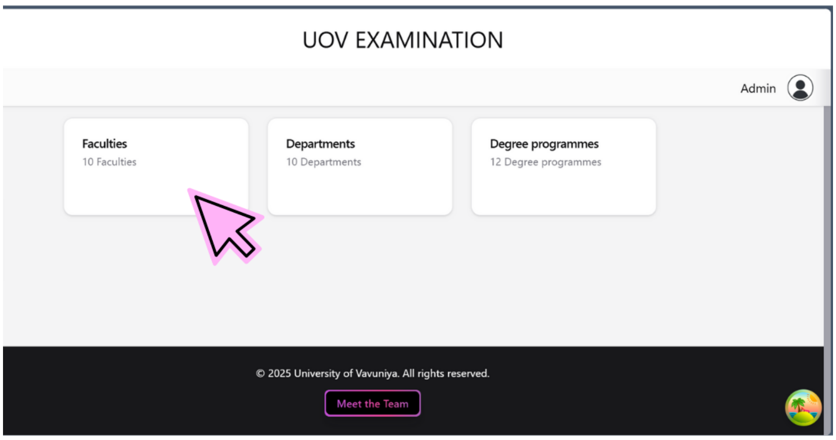

Figure 5.11: Courses

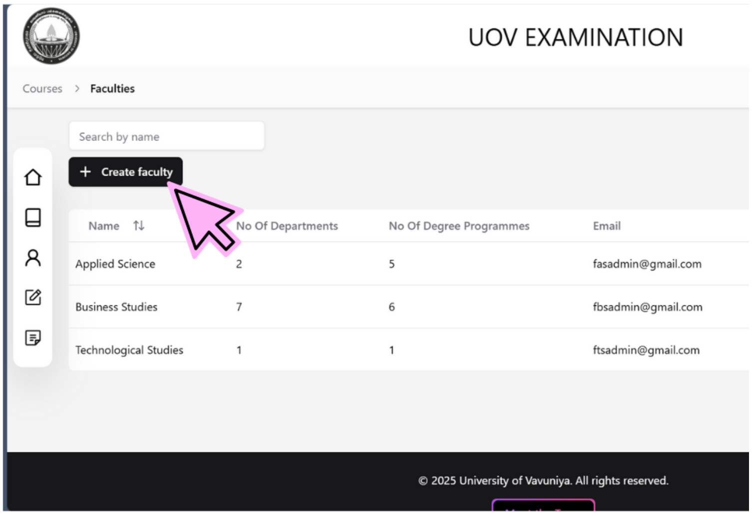

Figure 5.12: create faculty

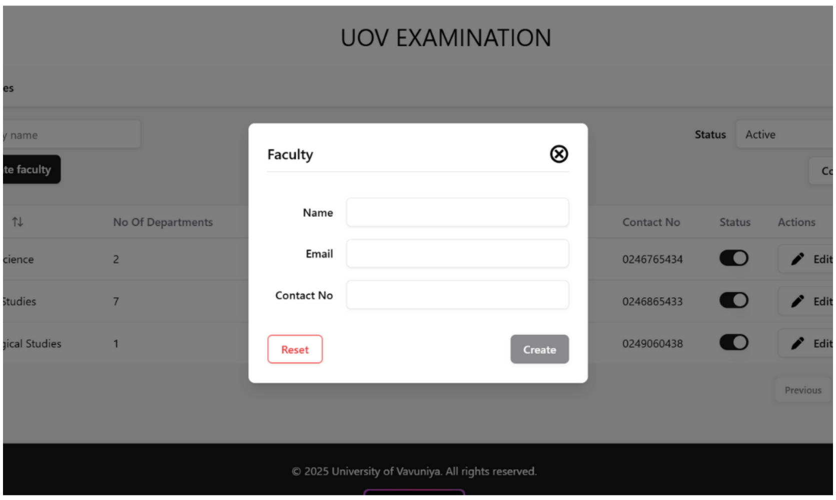

Figure 5.13: Creating process

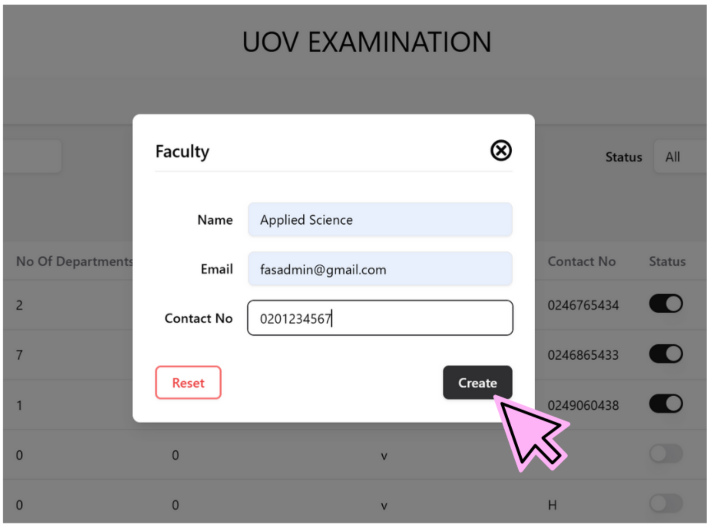

Figure 5.14: Create

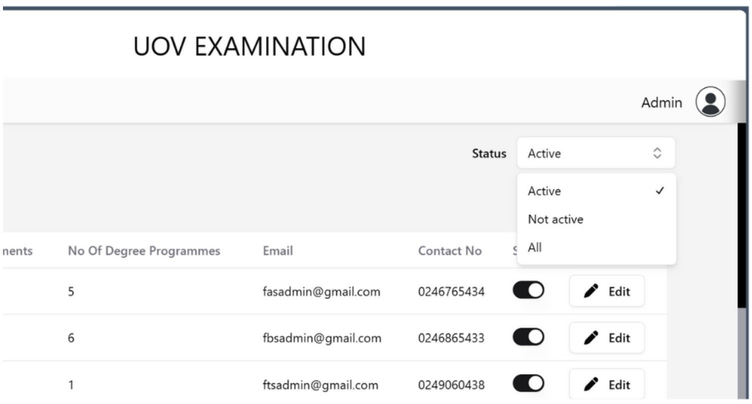

Figure 5.15: Edit status

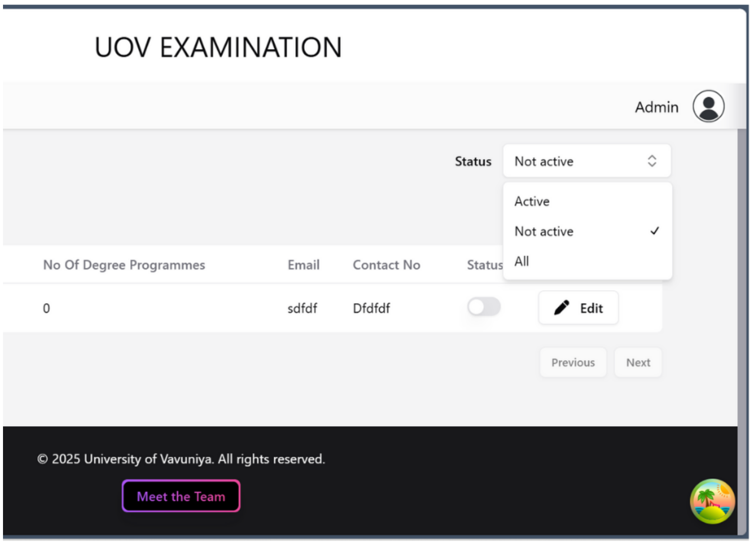

Figure 5.16: Status options

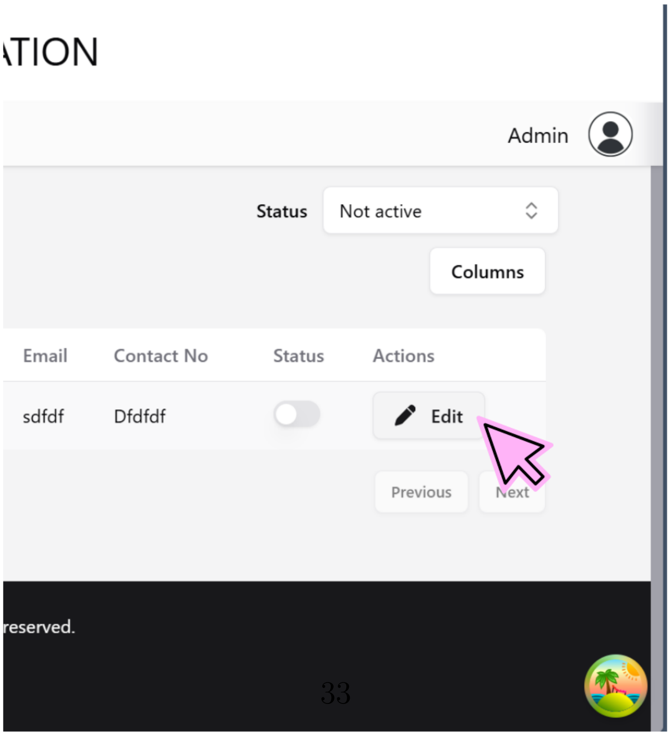

Figure 5.17: Edit

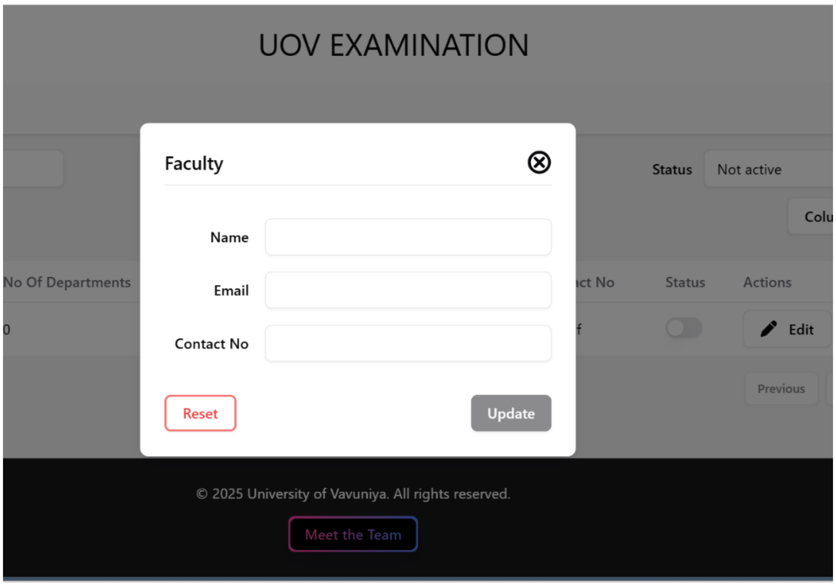

Figure 5.18: Update

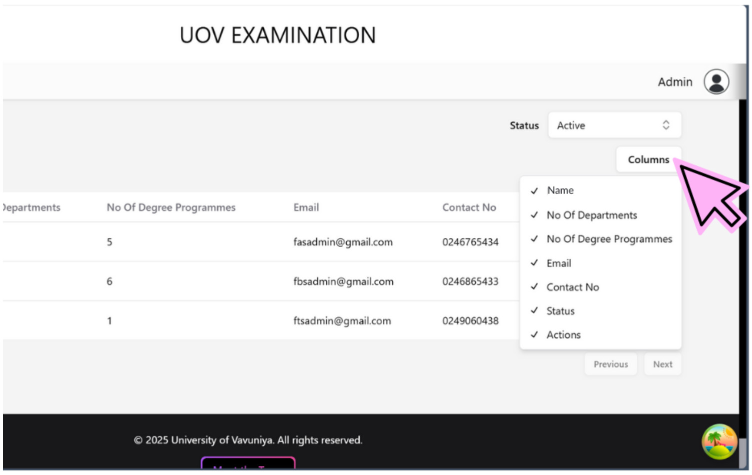

Figure 5.19: Display available columns

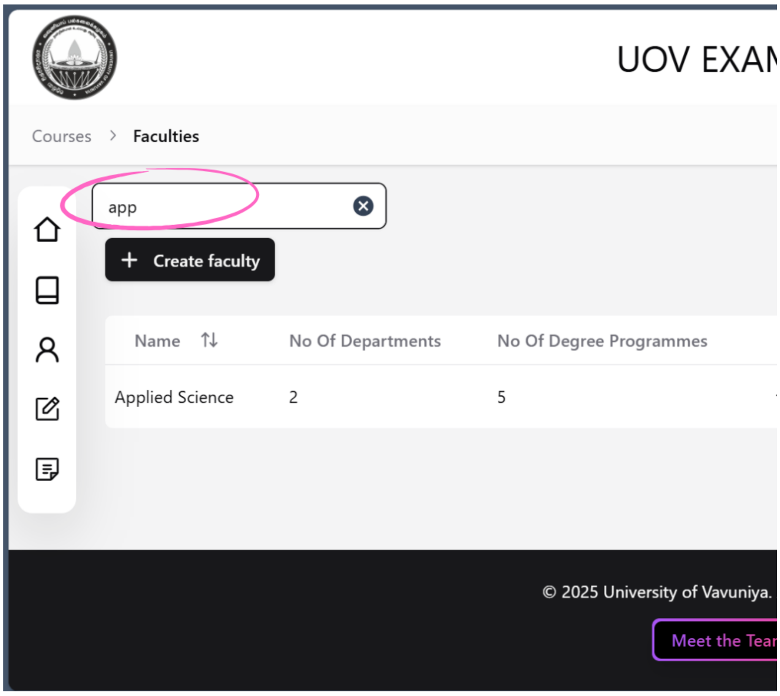

Figure 5.20: Search bar

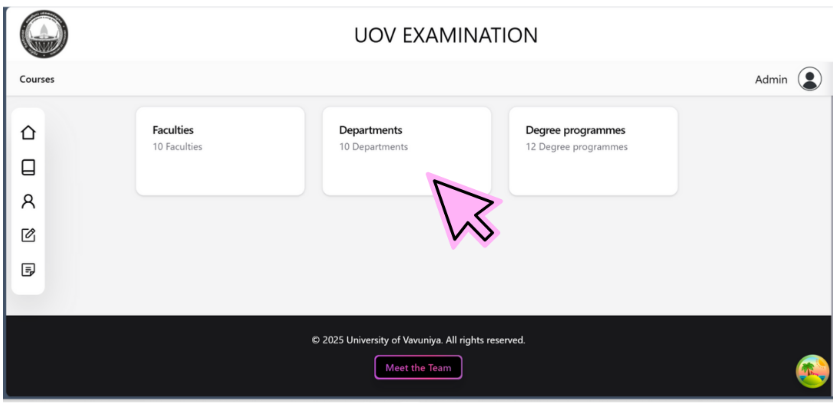

Figure 5.21: Departments

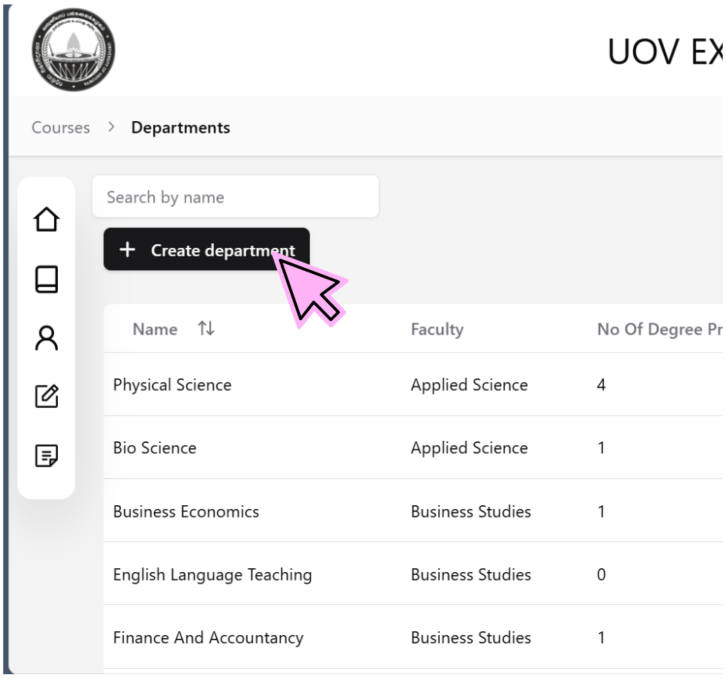

Figure 5.22: Create department

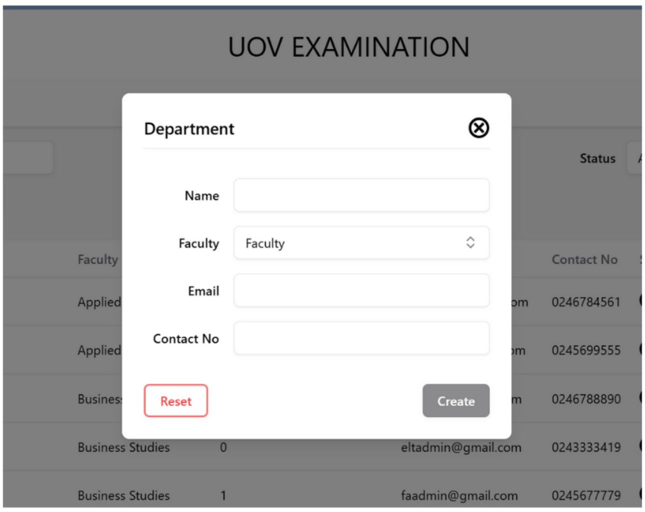

Figure 5.23: Creating process

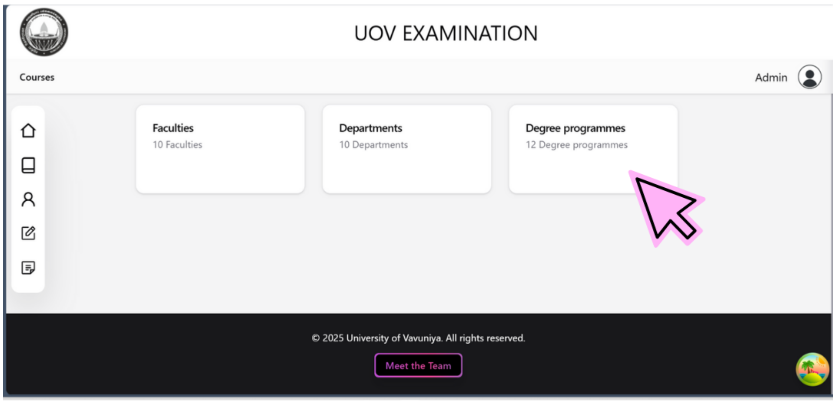

Figure 5.24: Degree Programmes

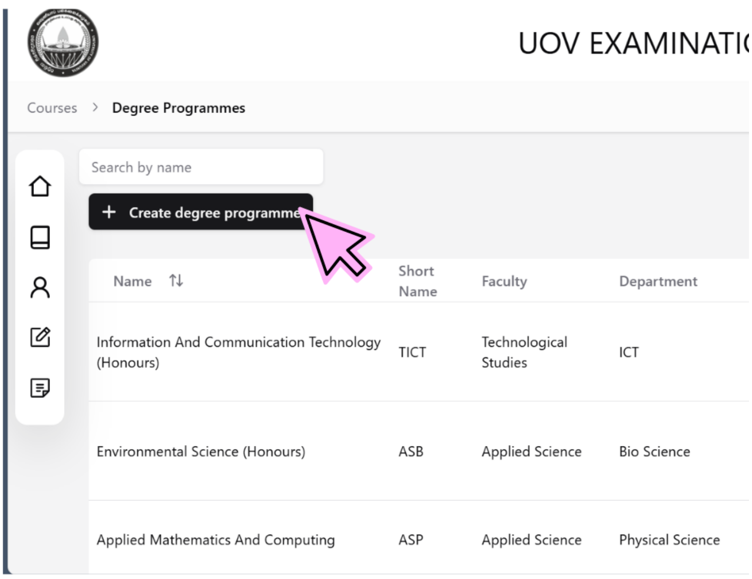

Figure 5.25: Create Degree Programmes

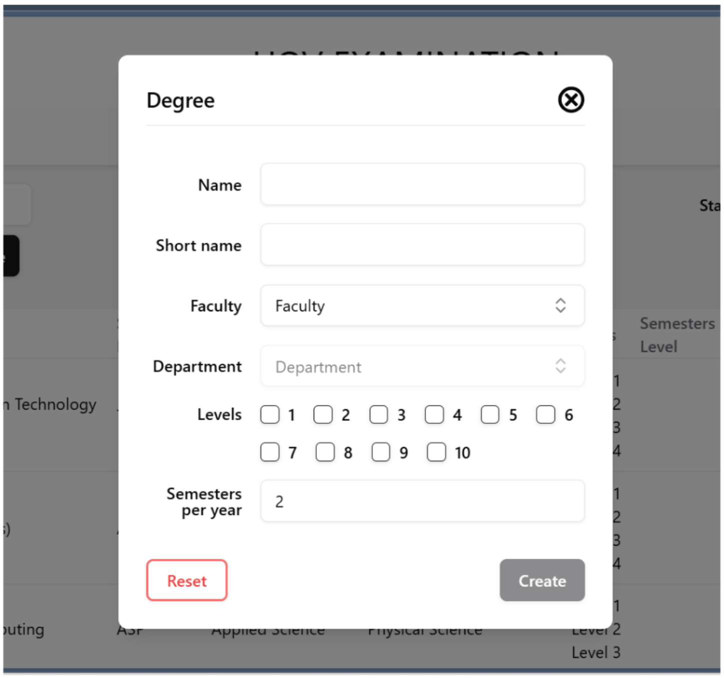

Figure 5.26: Creating process

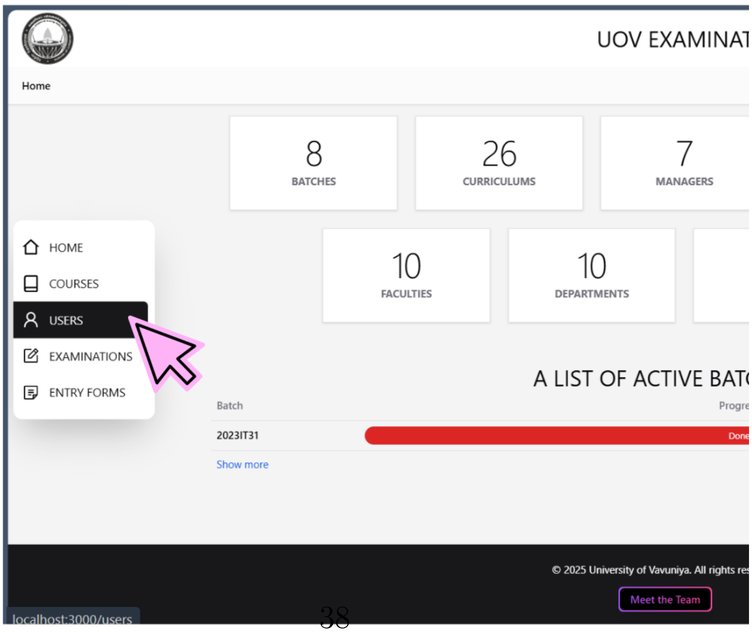

Figure 5.27: Users

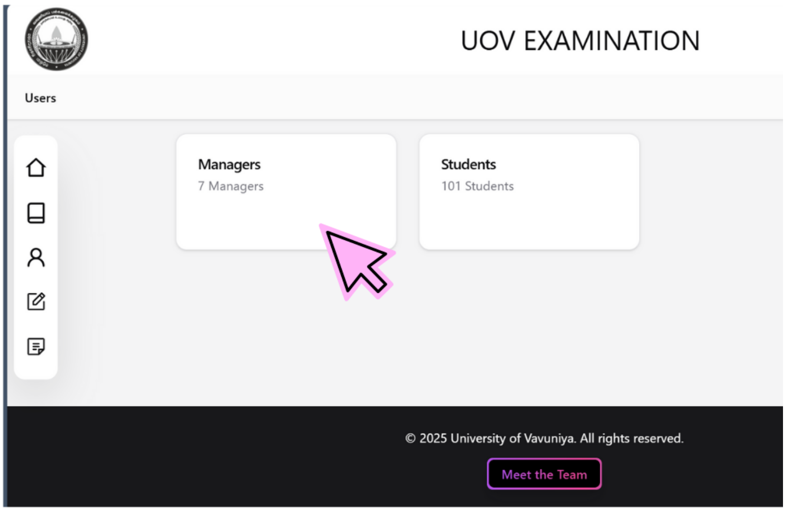

Figure 5.28: Users page

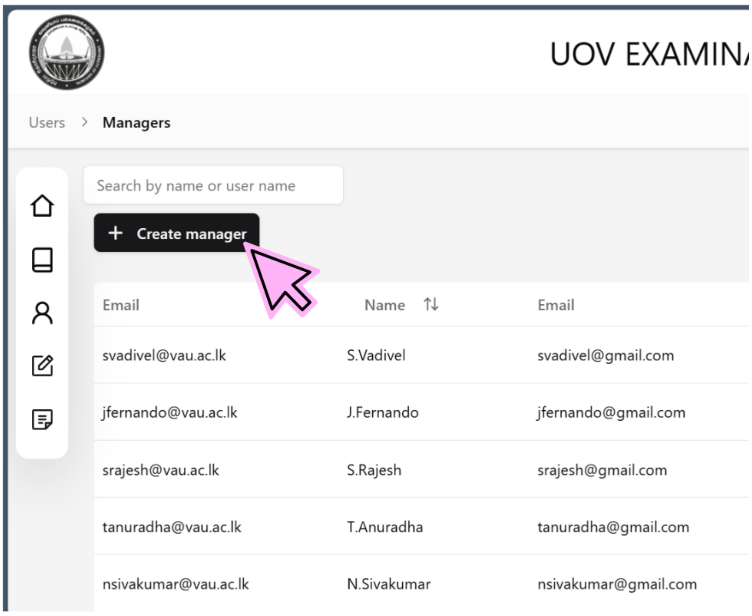

Figure 5.29: create manager

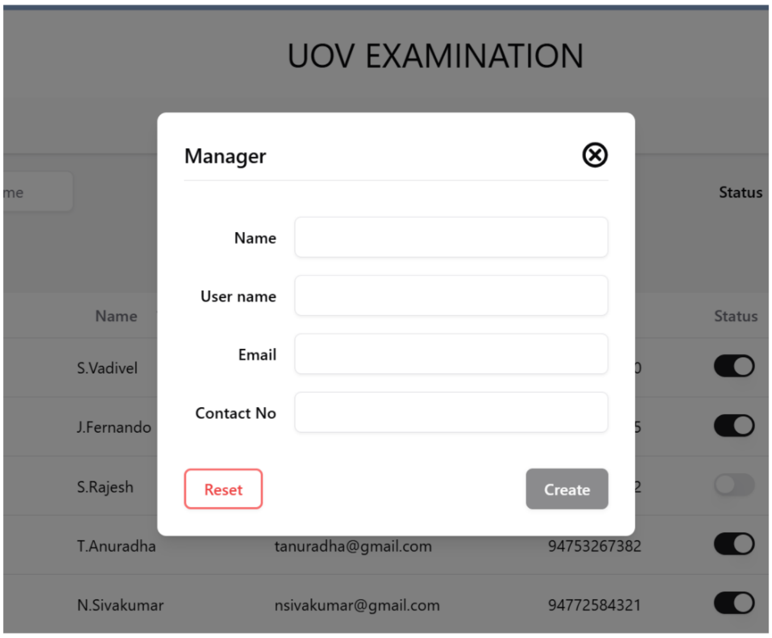

Figure 5.30: creating process

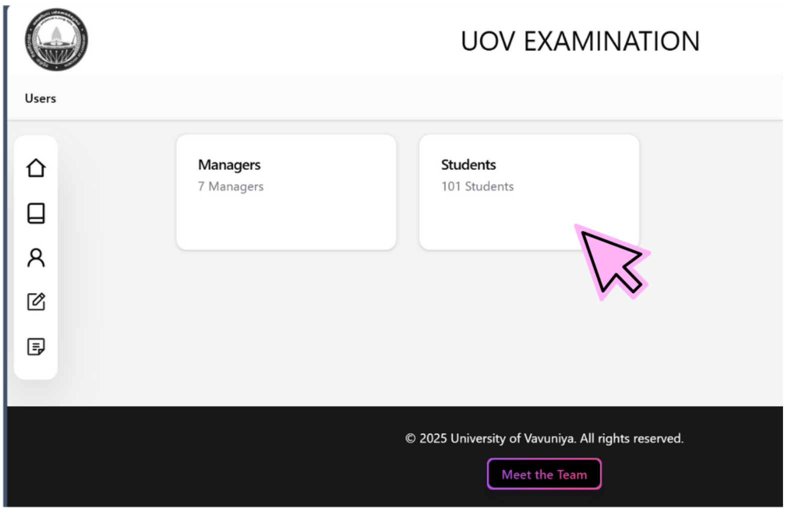

Figure 5.31: Click students

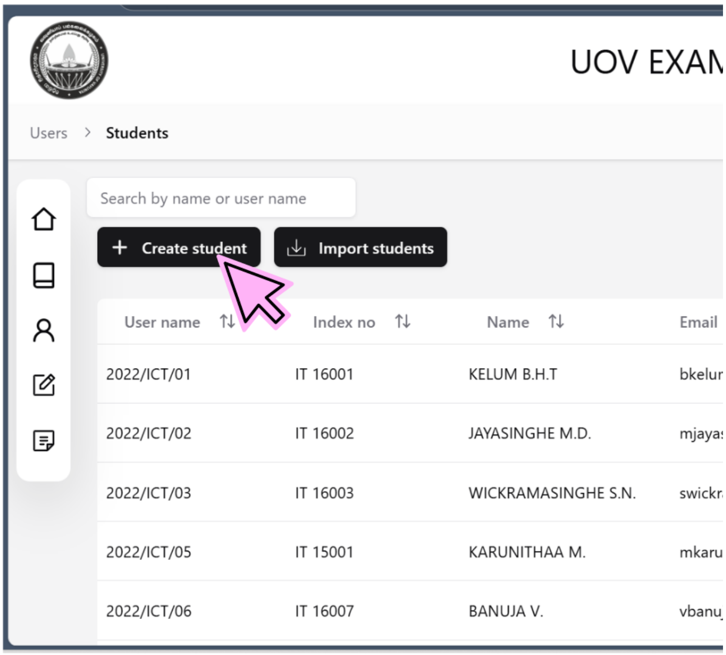

Figure 5.32: Create student

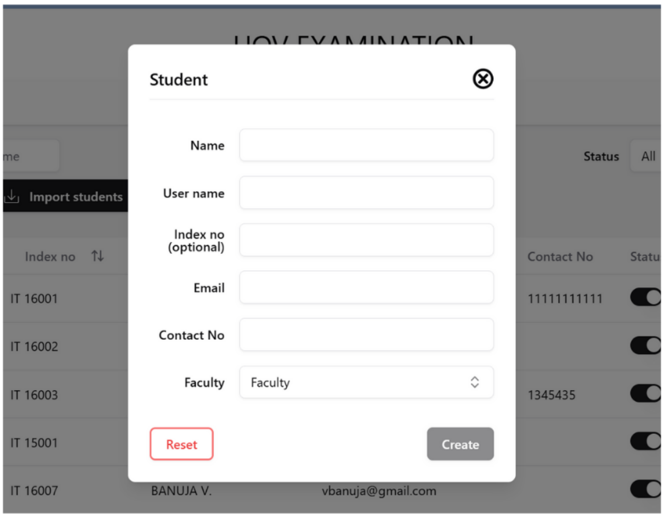

Figure 5.33: Creating process

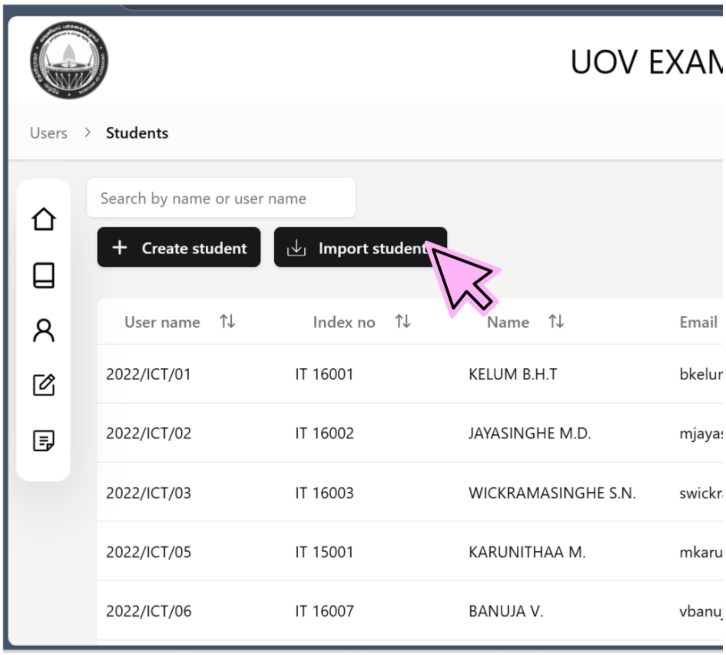

Figure 5.34: Import student

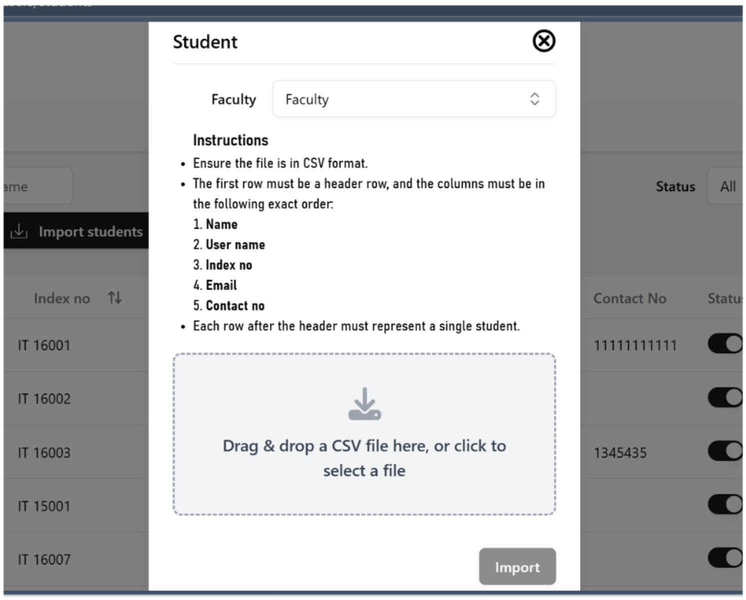

Figure 5.35: Importing process

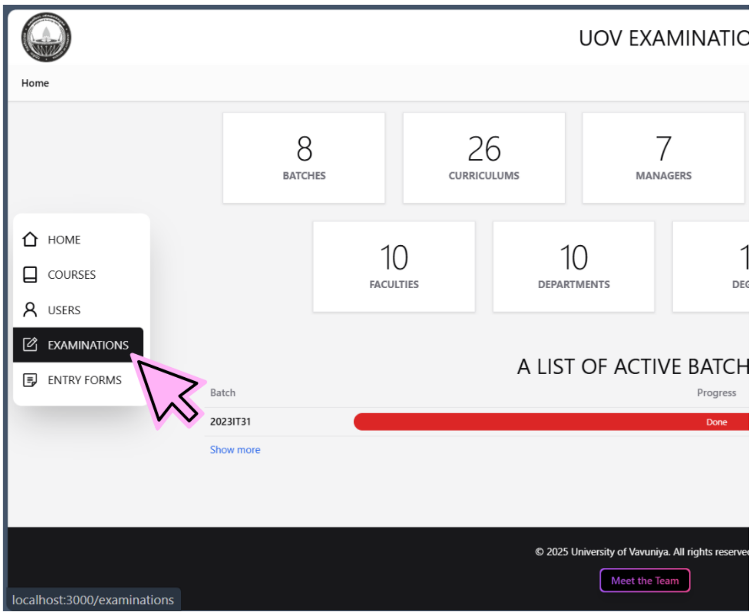

Figure 5.36: Examinations

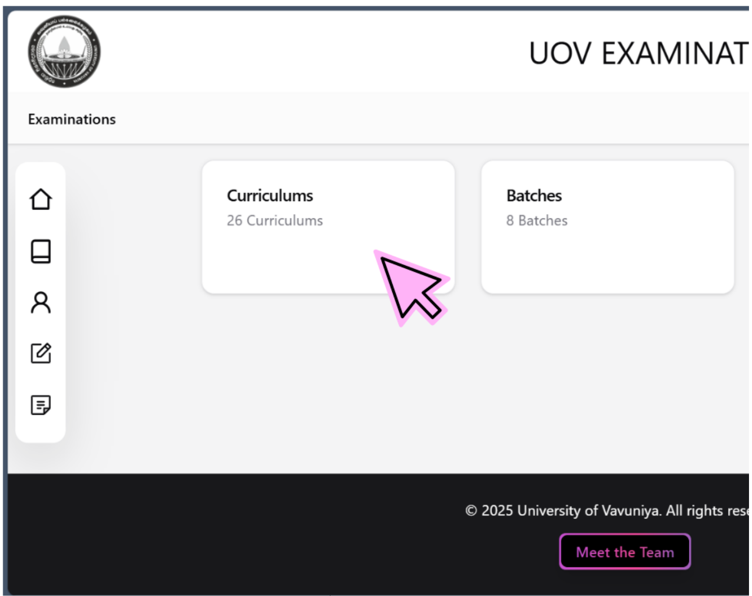

Figure 5.37: Examinations page

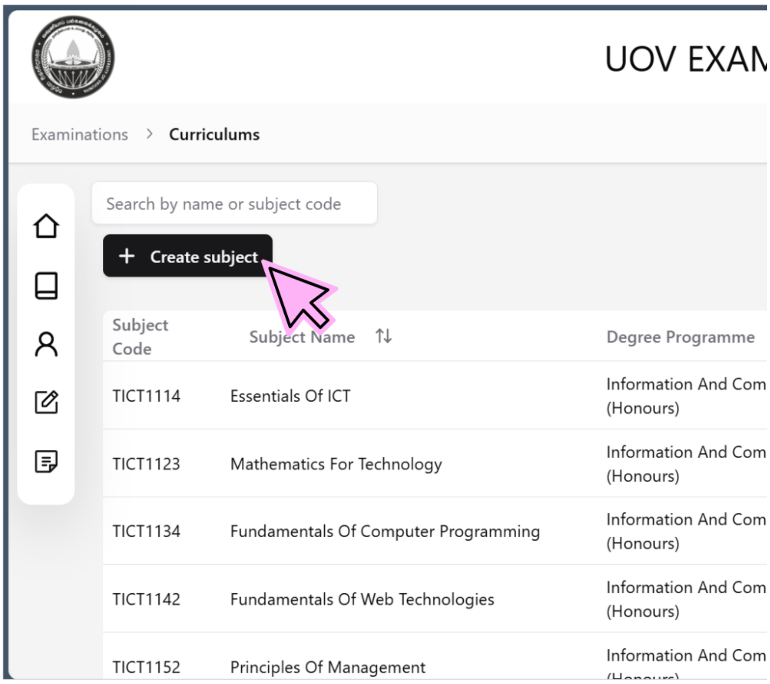

Figure 5.38: create subject

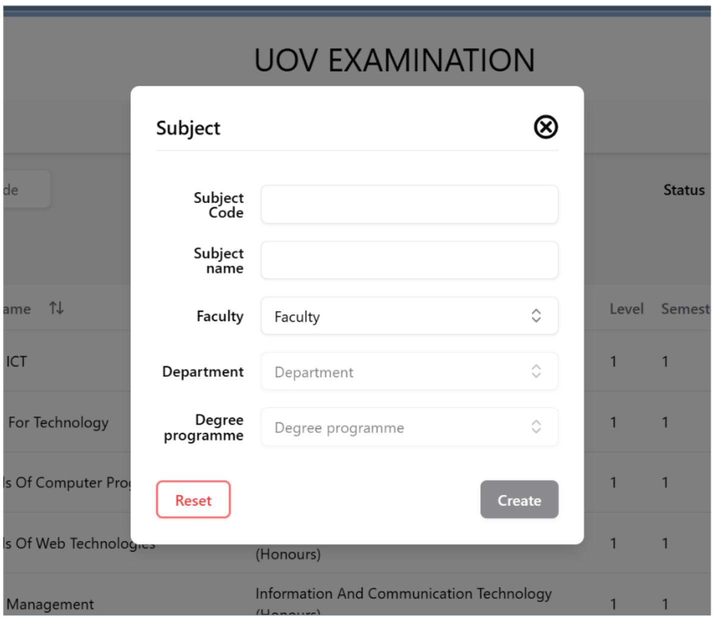

Figure 5.39: creating process

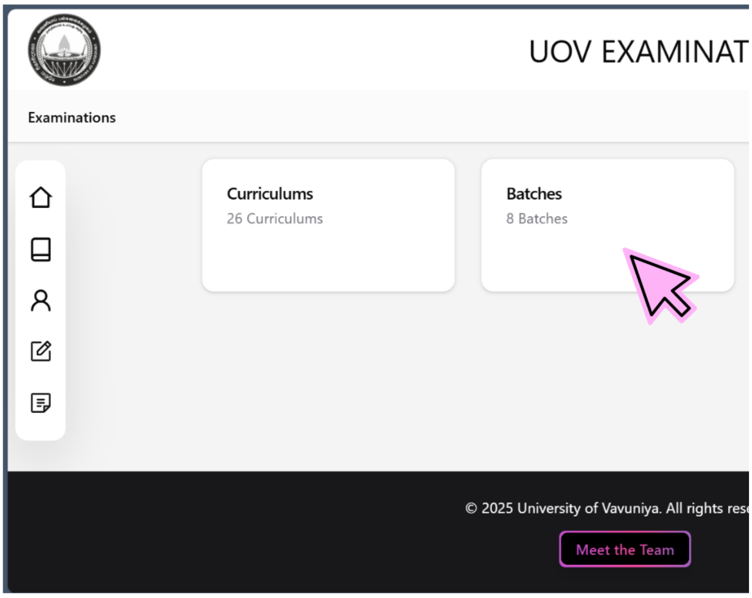

Figure 5.40: Batches

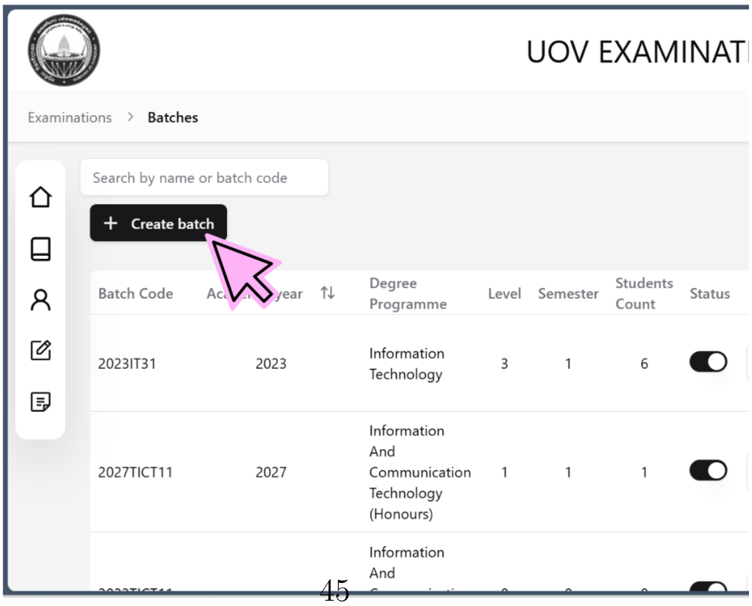

Figure 5.41: Create batch

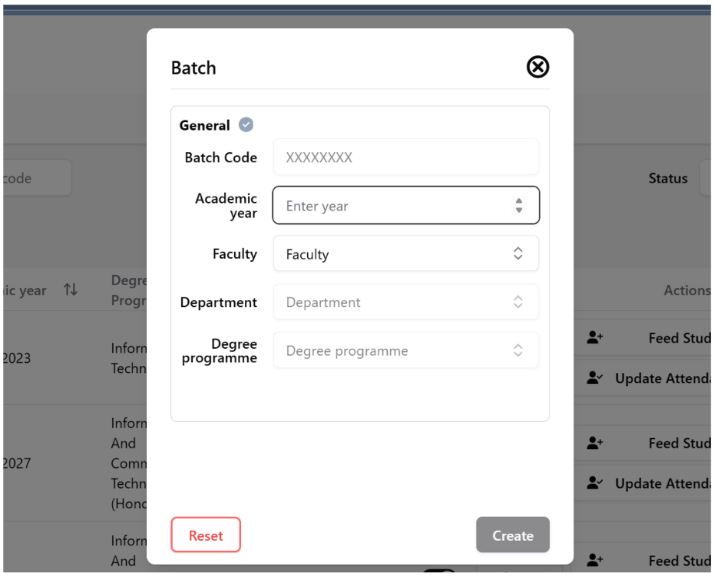

Figure 5.42: Creating process

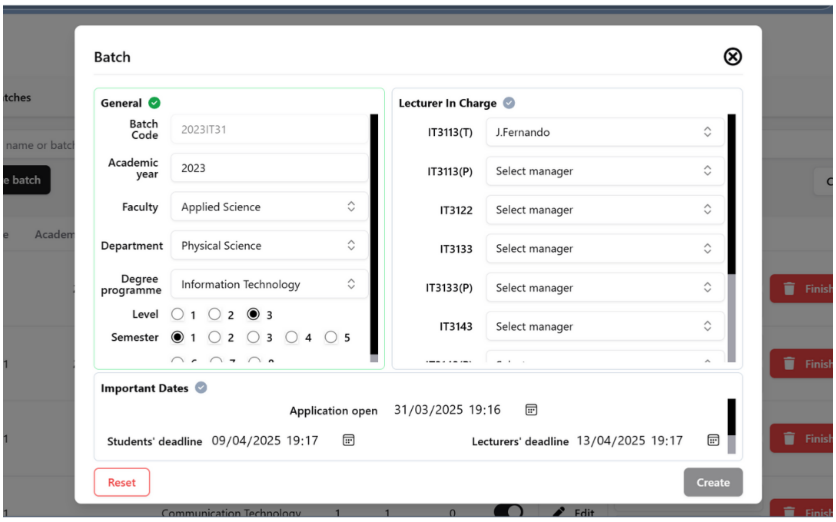

Figure 5.43: Creating process

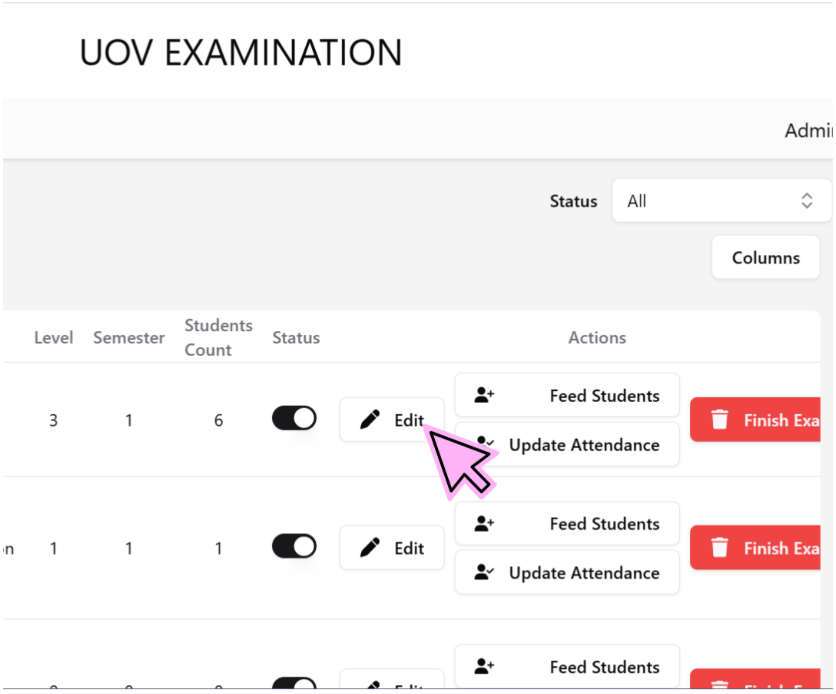

Figure 5.44: Edit options

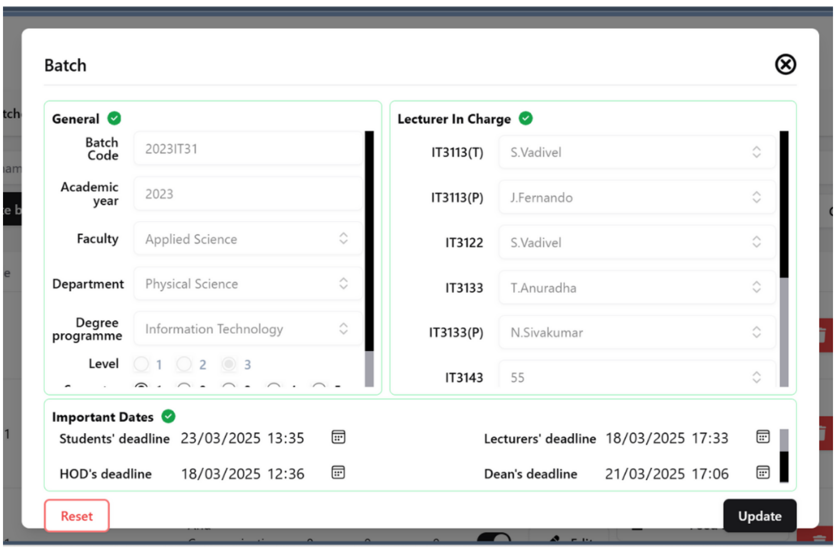

Figure 5.45: Update batch

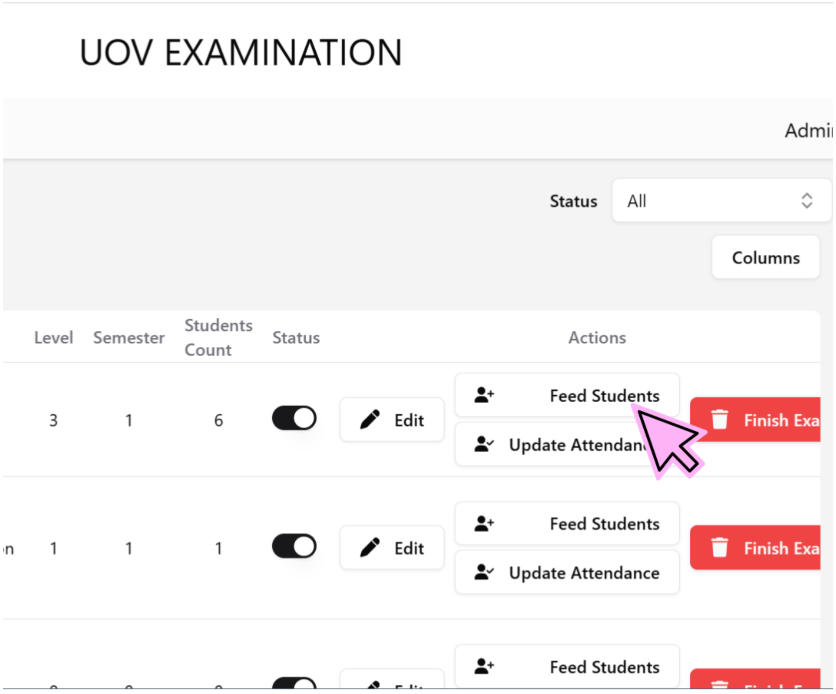

Figure 5.46: Feed Students

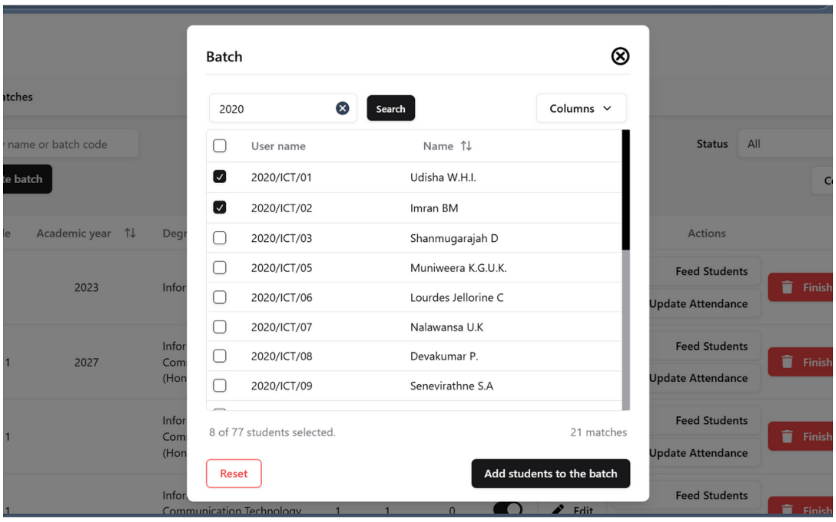

Figure 5.47: Process

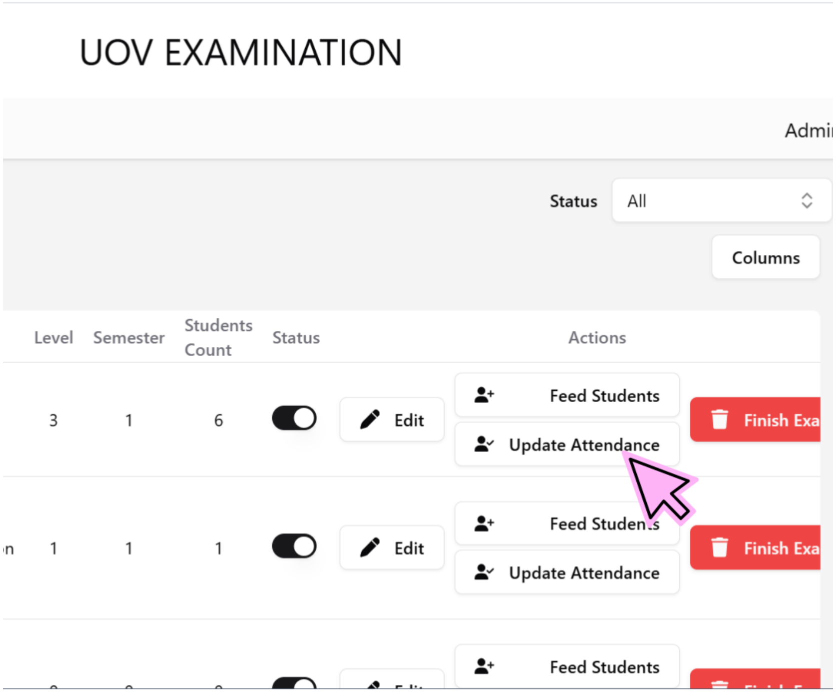

Figure 5.48: Update Attendance

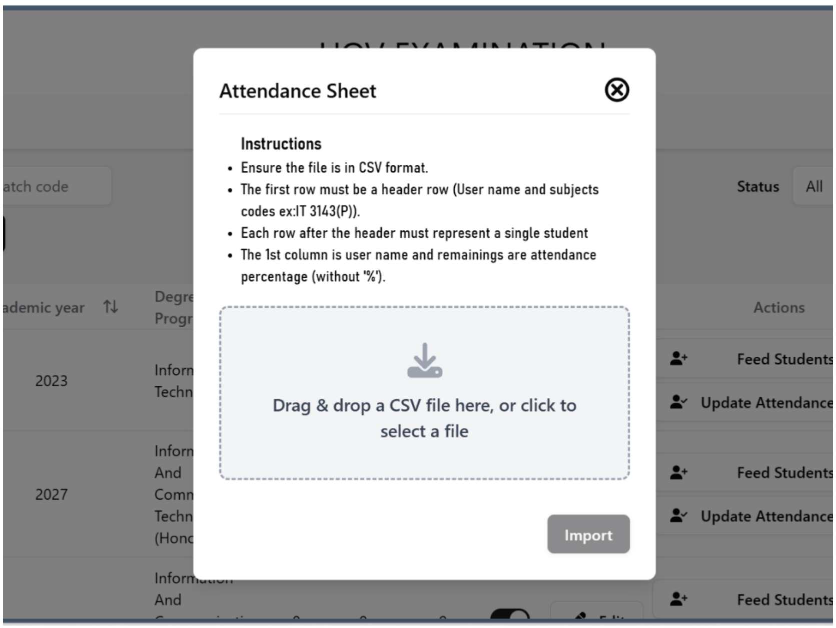

Figure 5.49: Updating process

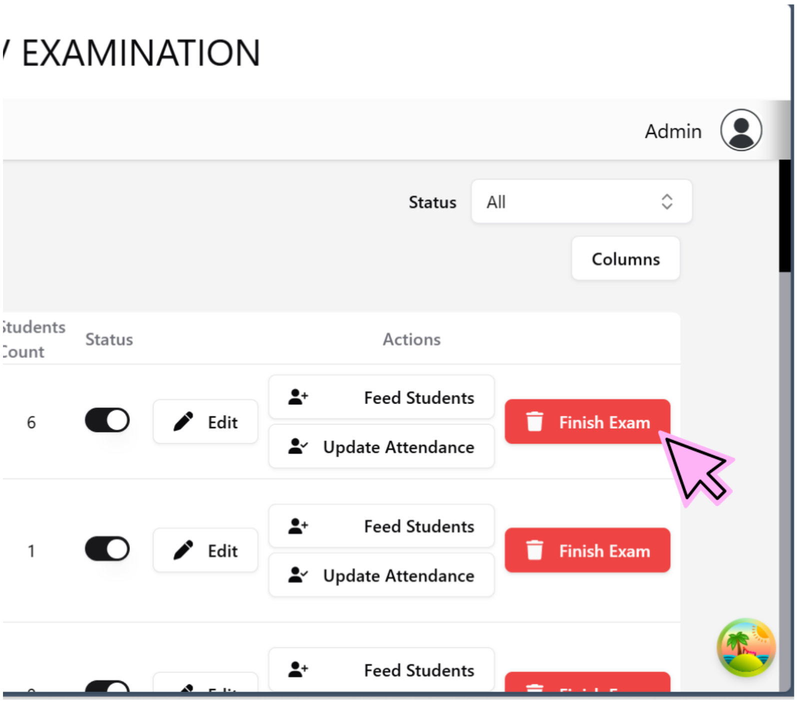

Figure 5.50: Finish Exam

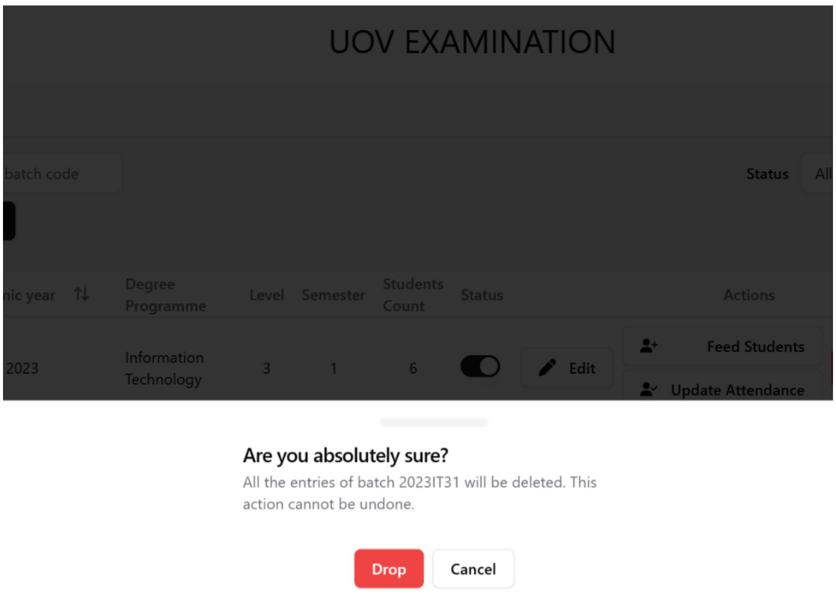

Figure 5.51: Drop

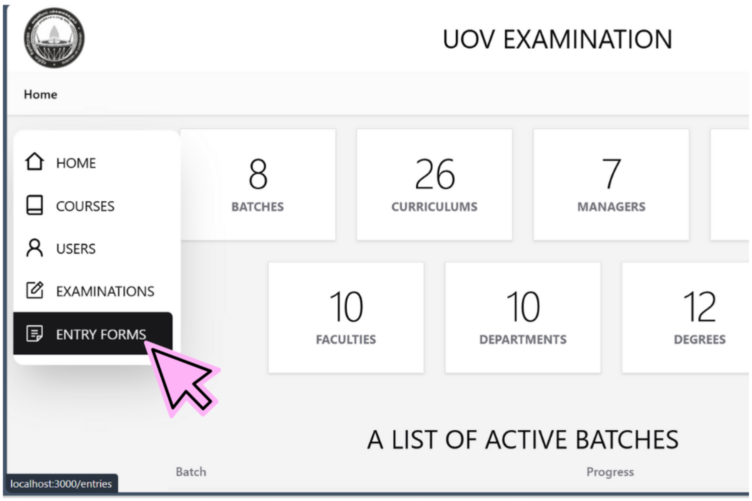

Figure 5.52: Entry Forms

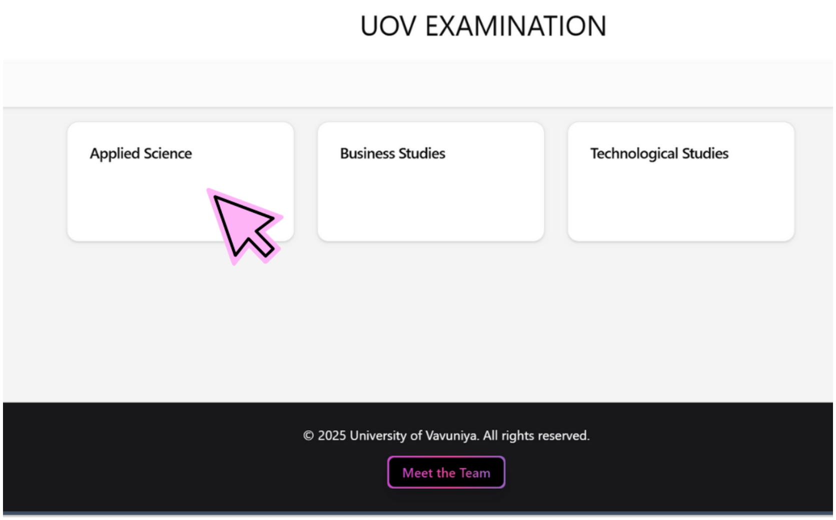

Figure 5.53: Entry Forms page

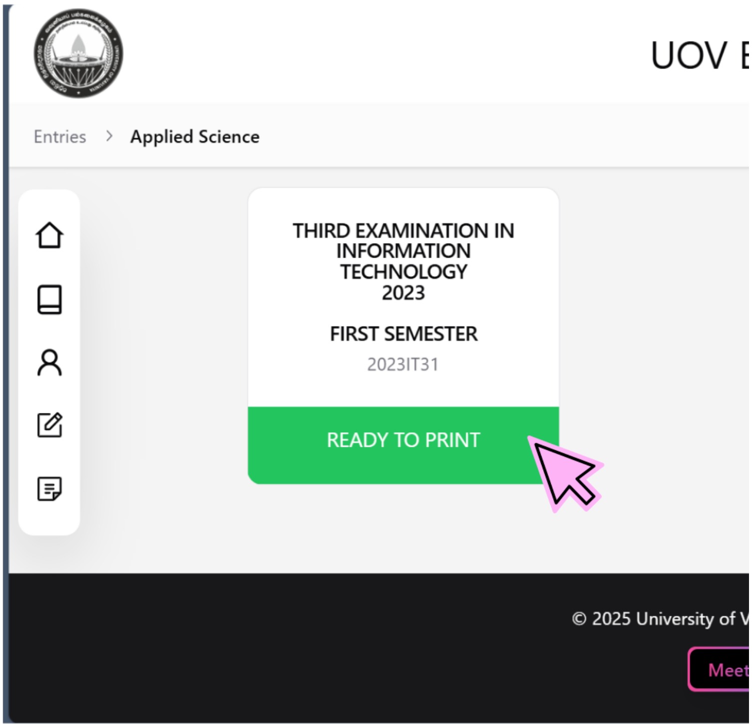

Figure 5.54: Ready to print

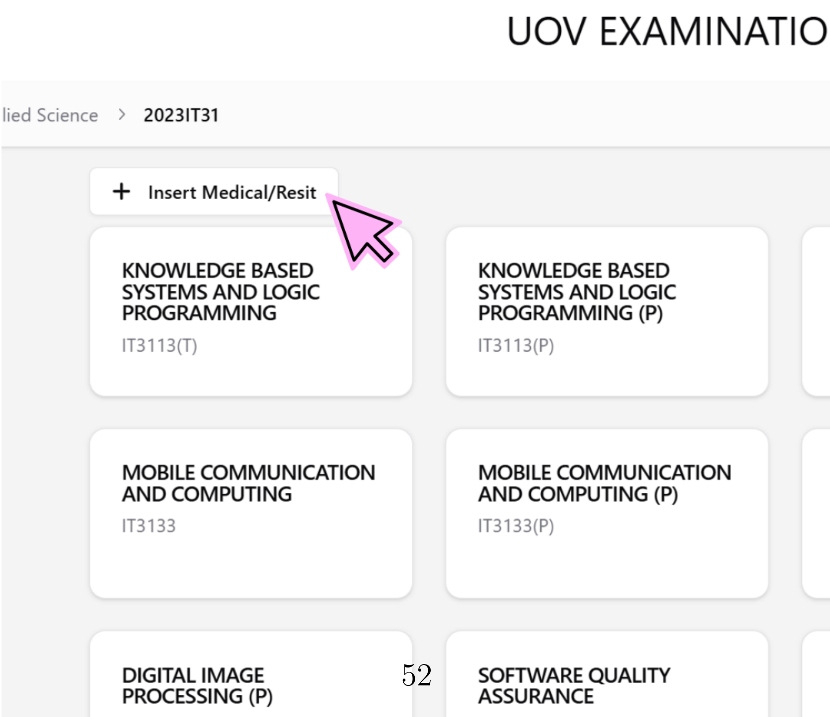

Figure 5.55: Insert Medical/Resit

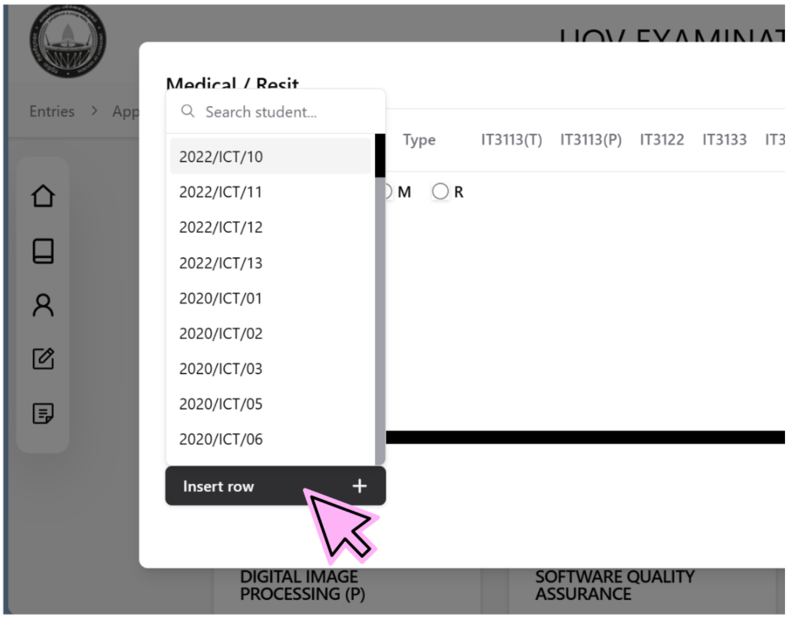

Figure 5.56: Insert row

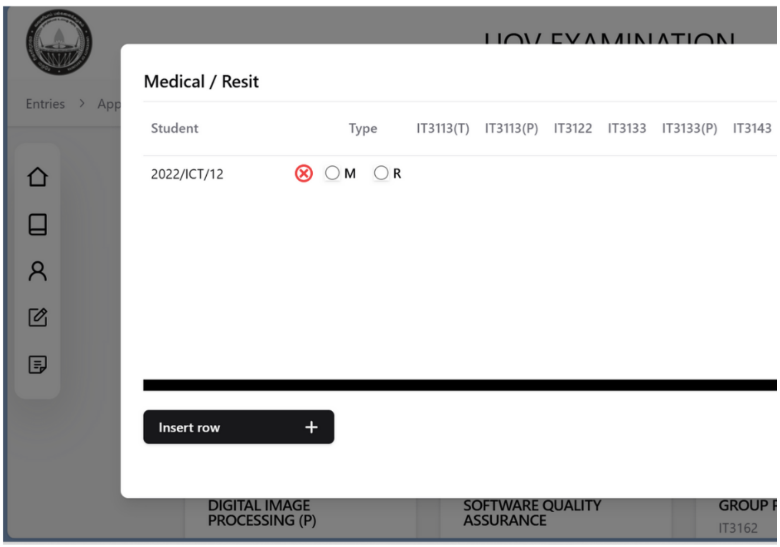

Figure 5.57: process

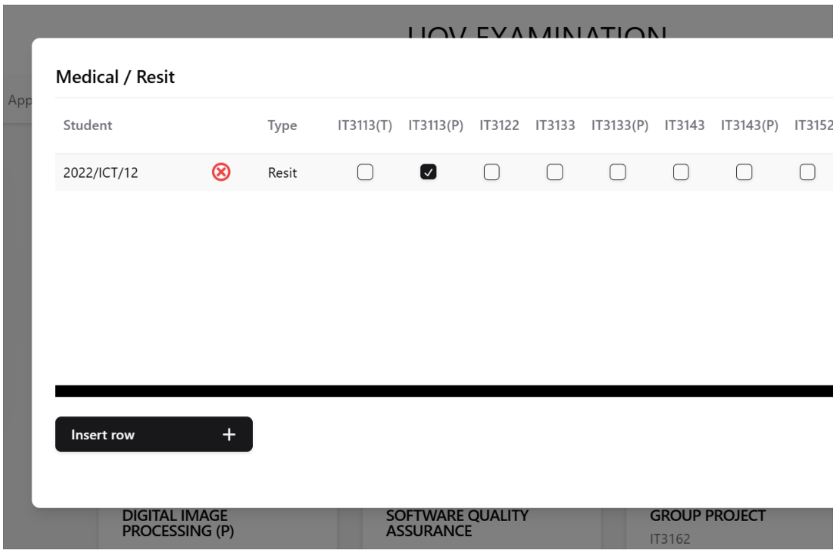

Figure 5.58: process

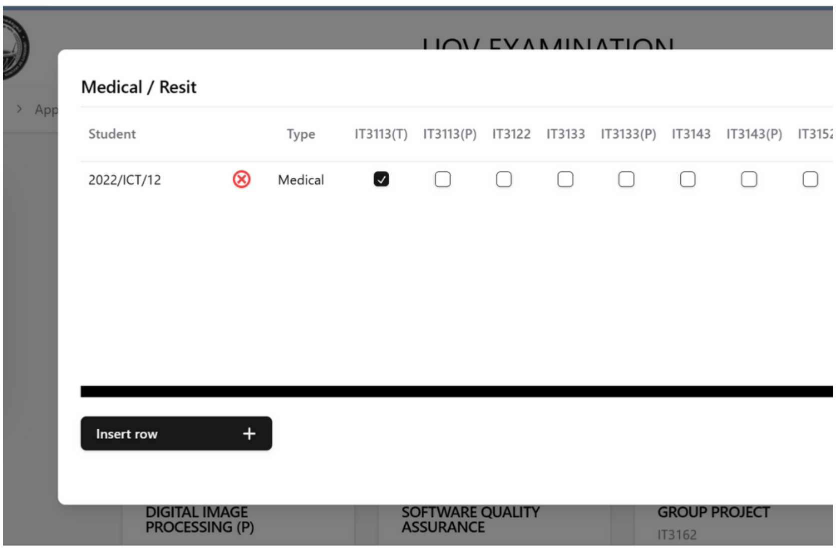

Figure 5.59: process

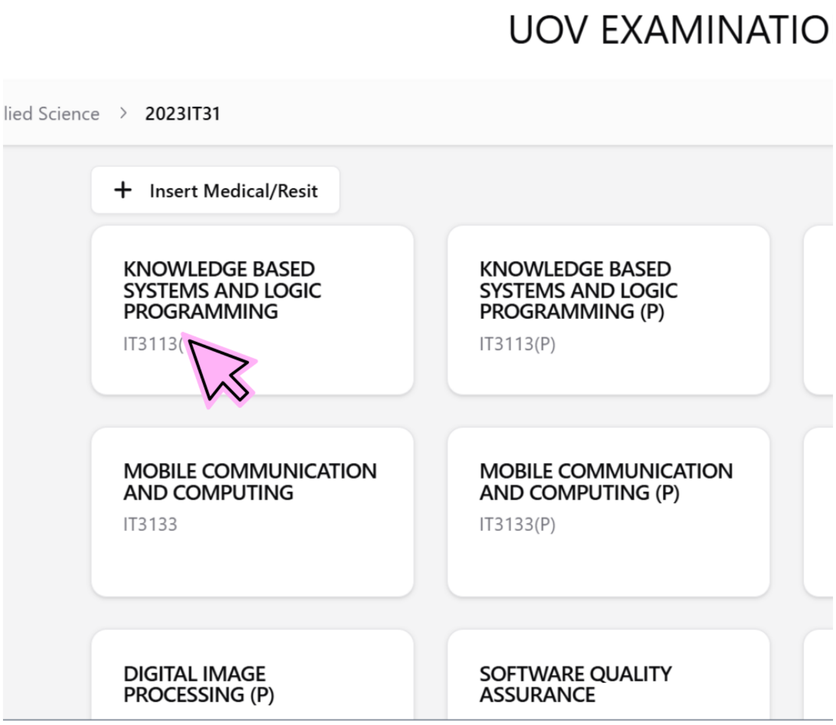

Figure 5.60: Add subjects

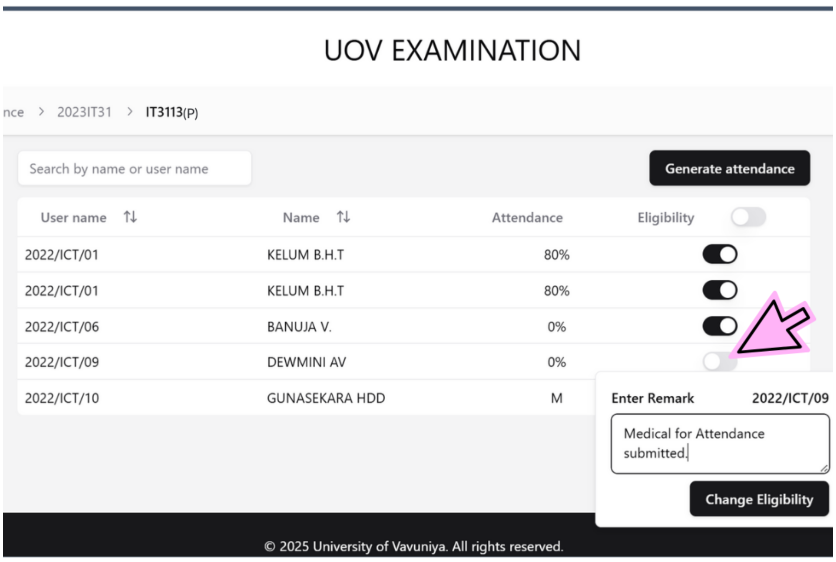

Figure 5.61: Eligibility for exams

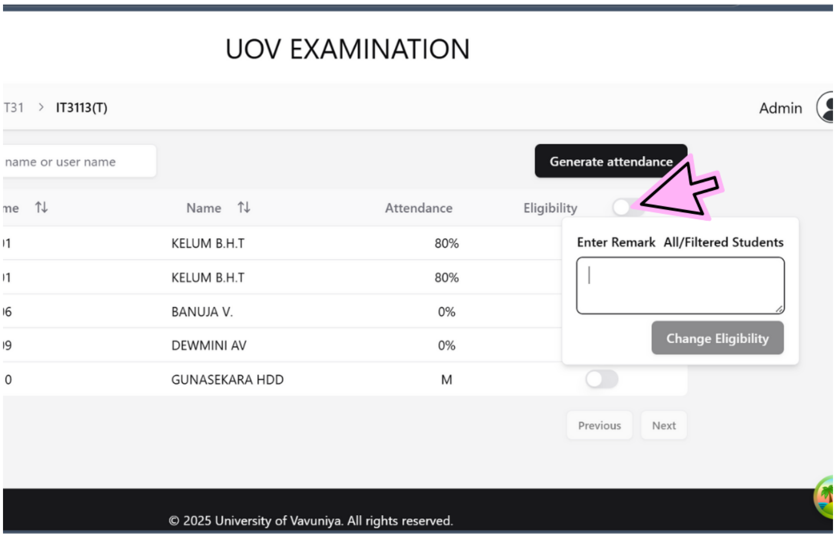

Figure 5.62: Process

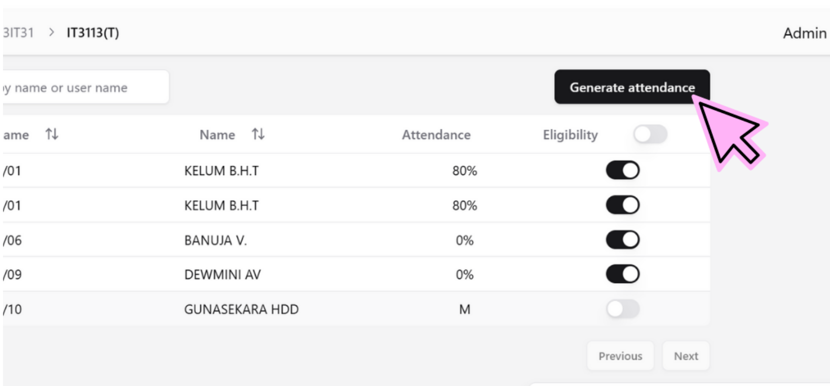

Figure 5.63: Generate Attendance

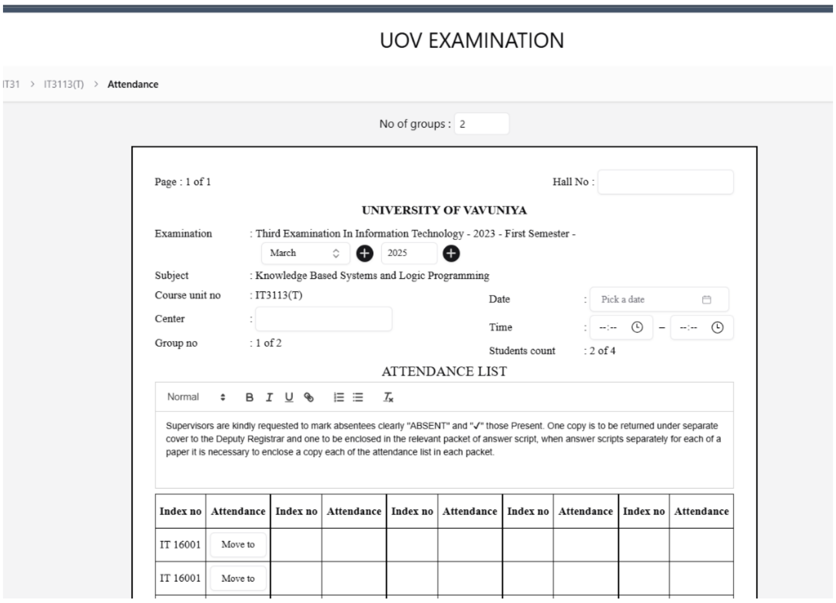

Figure 5.64: Filling the details

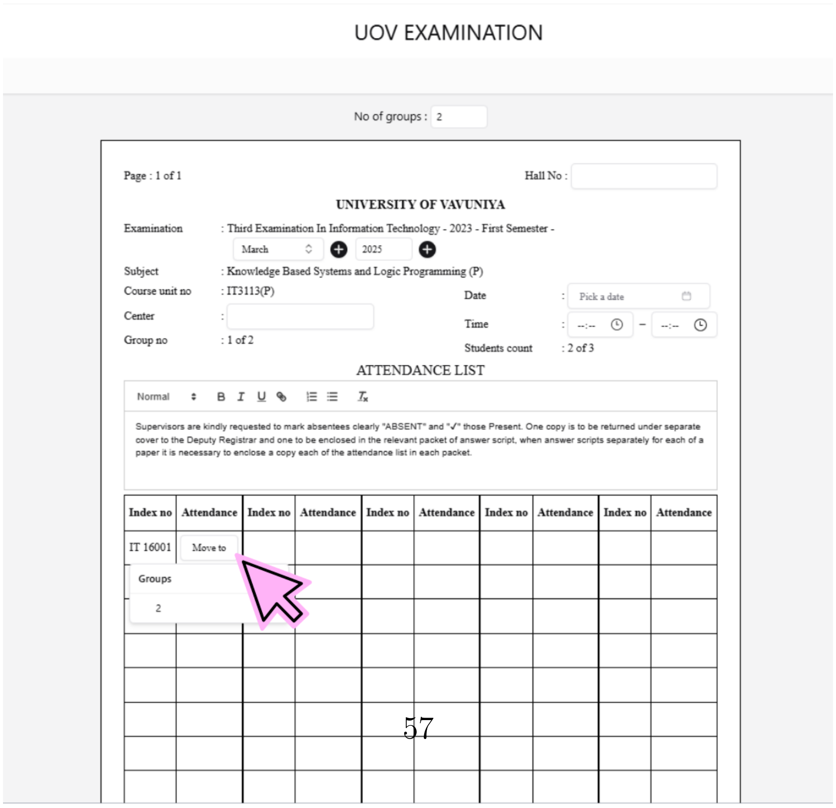

Figure 5.65: The Generate Attendance

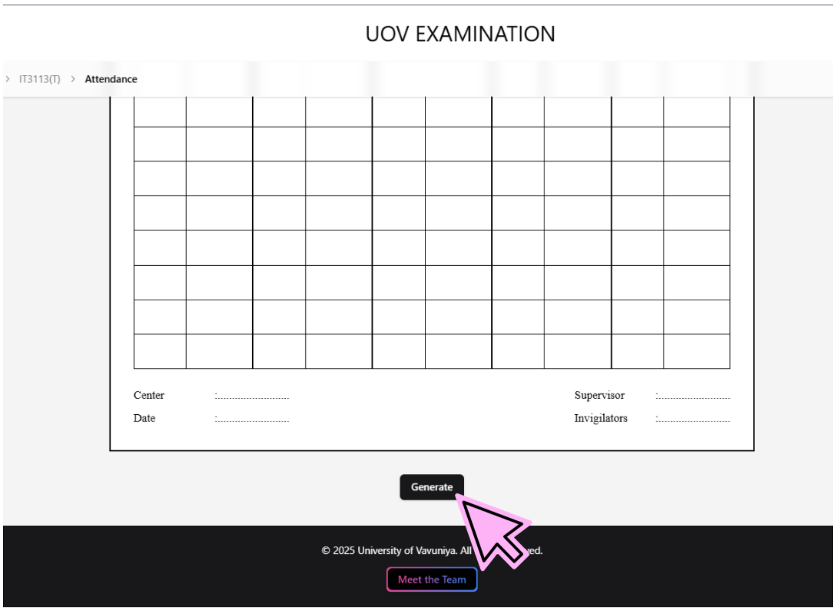

Figure 5.66: Generate

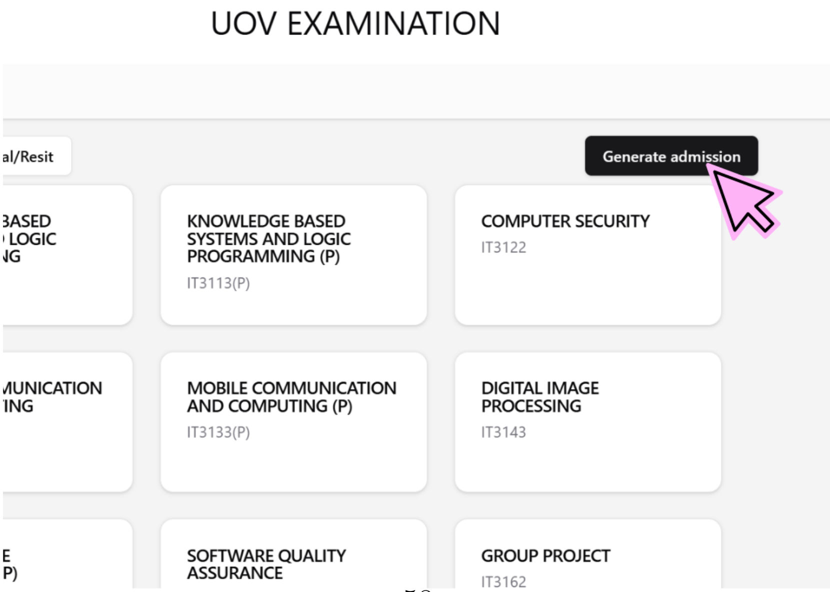

Figure 5.67: Generate Admission

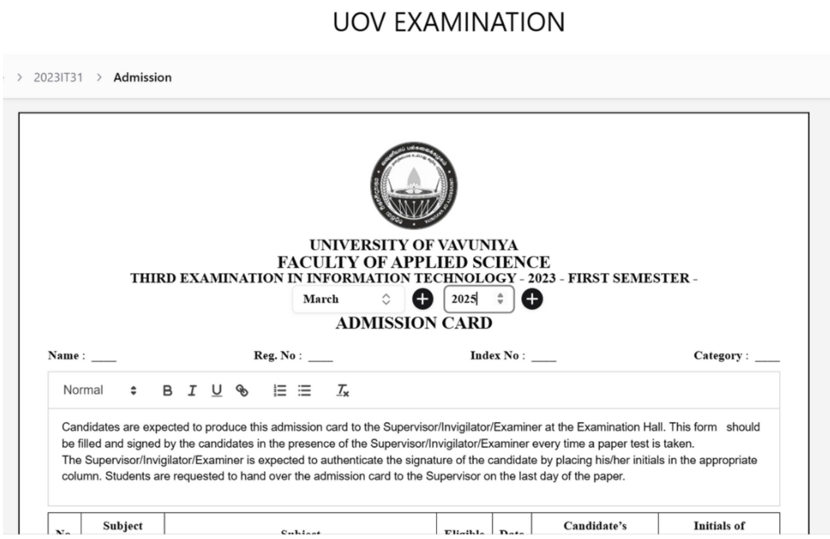

Figure 5.68: Editing

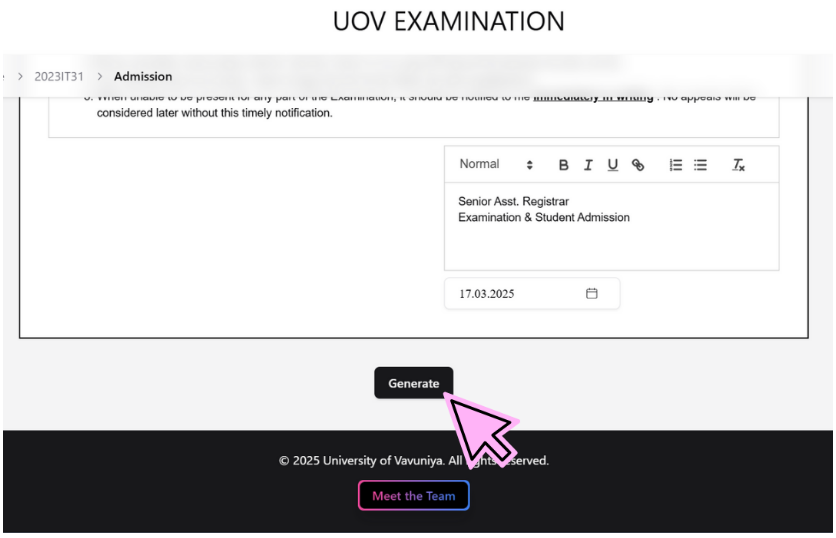

Figure 5.69: Generate

- Student Module

The Sign in shown in the Figure 5.70. The student can log in using the correct username and password.

The View eligibility status shown in the Figure 5.71. After applying for an examination, the student can view their eligibility status.

The Download admission card. shown in the Figure 5.73. After submitting the required information, the student can download their admission card.

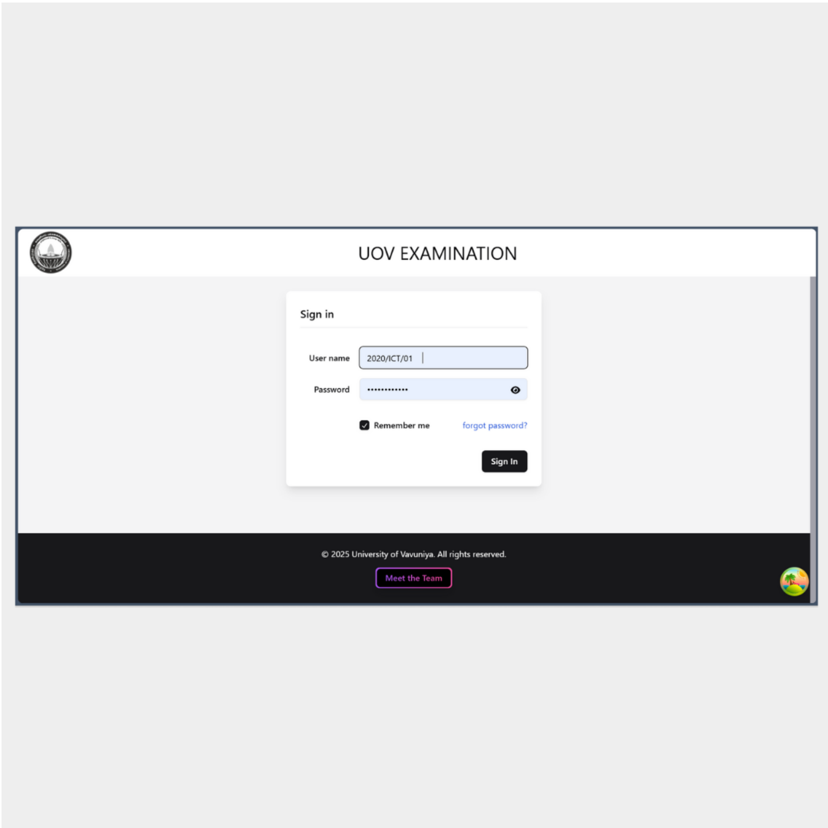

Figure 5.70: Sign in

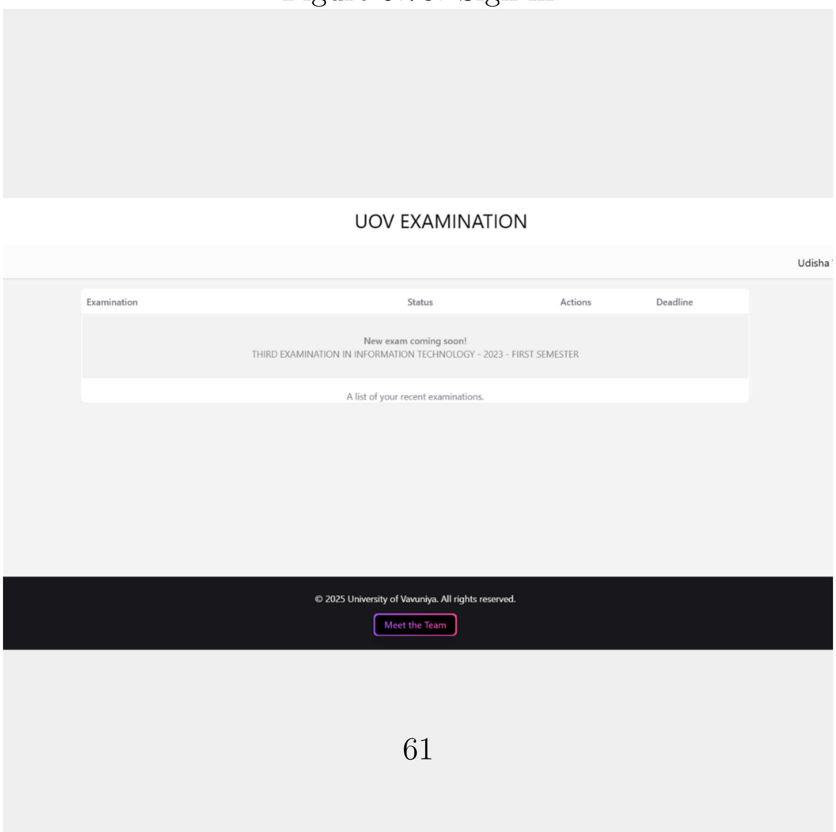

Figure 5.71: View eligibility status

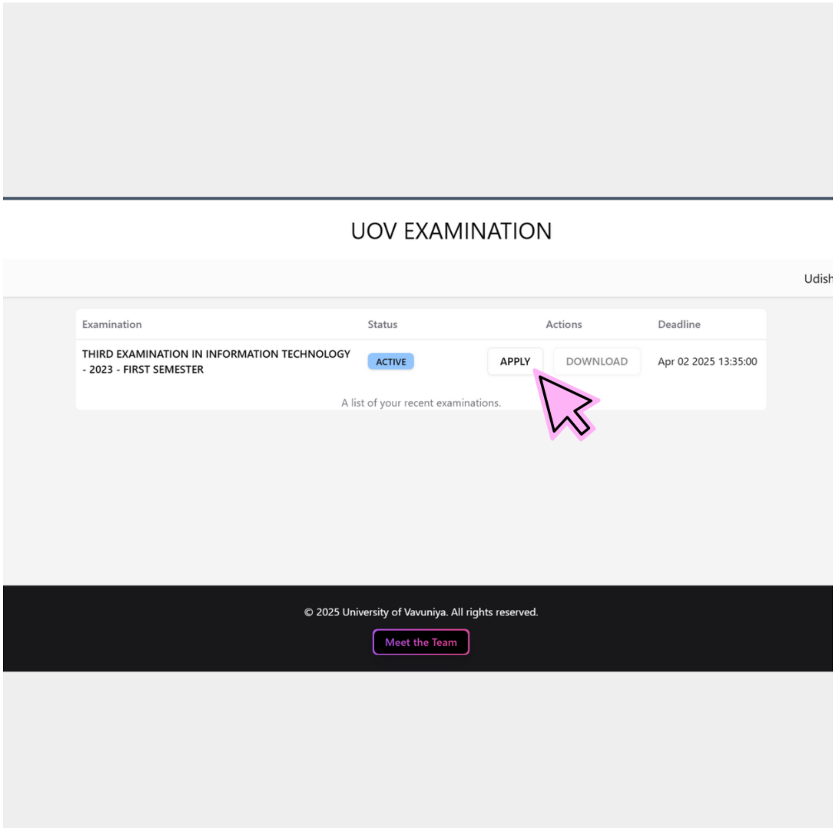

Figure 5.72: Applying part

Figure 5.73: Download admission card

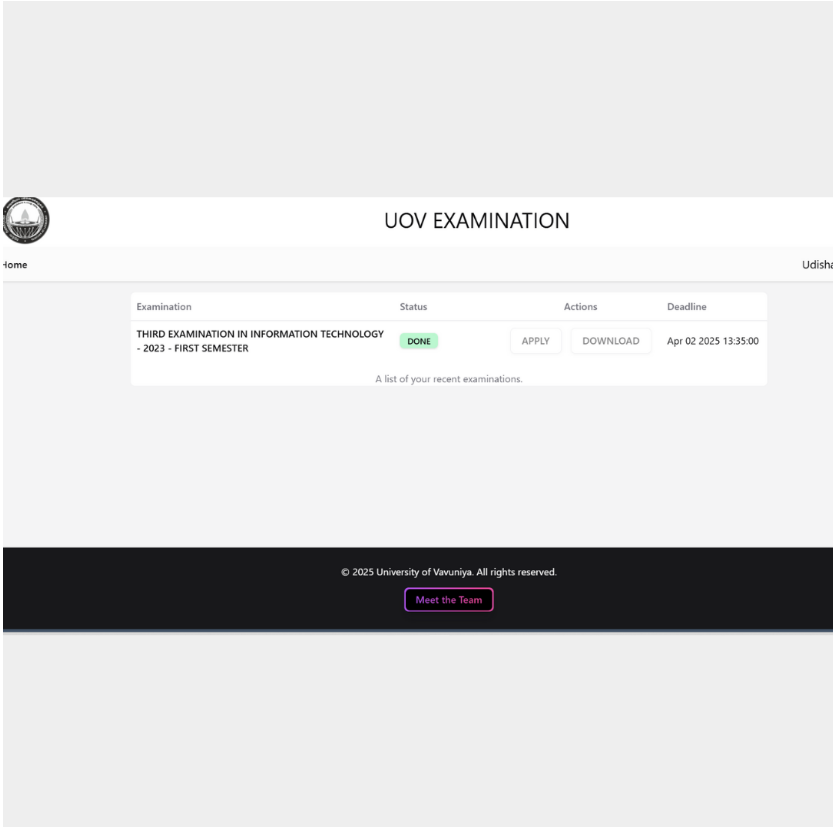

Figure 5.74: Apply to download admission

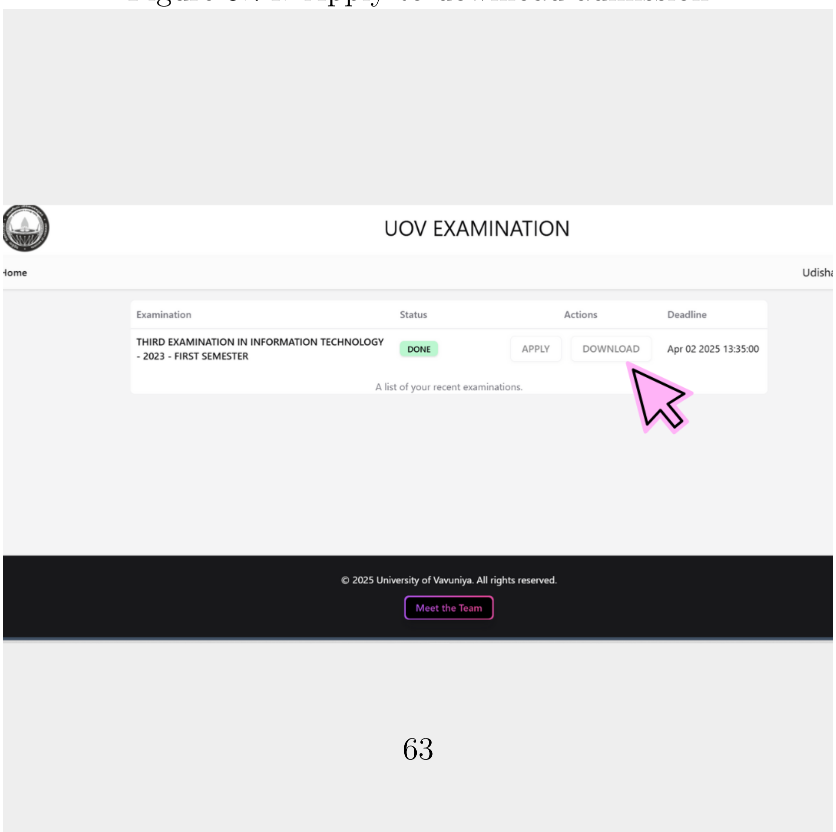

Figure 5.75: Download admission

Chapter 6

Limitations, Recommendations and Conclusion

6.1 Introduction

The Examination Entry System (EES) has been designed to streamline the exam registration and eligibility verification process within the university. By automating traditionally manual tasks, the system aims to enhance efficiency, accuracy, and transparency in exam management.

However, like any system, EES is not without its limitations. Certain technical constraints, user-related challenges, and system dependencies may impact its overall performance. Addressing these limitations is essential to ensure the system operates at its full potential.

To improve its functionality and effectiveness, several recommendations can be made, focusing on enhancing security, optimizing system performance, and improving user experience. Implementing these improvements will help overcome existing challenges and ensure long-term success.

While EES offers significant benefits in managing exam registrations efficiently, continuous evaluation and enhancement are required to make it a fully scalable, secure, and user-friendly system.

6.2 Limitations of the System

01.Technical Limitations

- Internet Dependency :– Since the system is web-based, students and staff must have internet access to use it.
- System Downtime Risks :– If the server goes down or undergoes maintenance, users may be unable to access the platform.
- Scalability Concerns :– The current system may not handle a large number of concurrent users efficiently if demand increases significantly.Data Security Challenges – Although the system implements role-based access control, unauthorized access or data leaks are still potential threats if not managed properly.Eligibility Verification
- Monitoring and Oversight
- Print Admissions Sheets

02.User-Related Limitations

- Learning Curve for Users :– Some students and staff may struggle with adapting to the new system, especially those with limited technical knowledge.
- Human Error in Verification :– Even though the system automates eligibility checking, errors in input data (e.g., attendance records) could lead to incorrect approvals or rejections.

6.3 Recommendations for Improvement

01.Technical Enhancements

- Implement an Offline Mode :– Allow students to fill in applications offline and sync them when they regain internet access.
- Optimize System Performance :– Improve server capabilities and implement load balancing to handle high traffic periods, especially before exam deadlines.
- Stronger Security Measures :– Use multi-factor authentication (MFA) for login, encryption for sensitive data, and automated security audits to detect vulnerabilities.

- Cloud-Based Backup :– Ensure automatic backups in case of system failures to prevent data loss.

02.User Experience Improvements

- User Training and Support :– Provide video tutorials and user guides for students and staff to minimize the learning curve.
- Automated Notifications :– Enable email/SMS alerts for application deadlines, approvals, and attendance status updates.

6.4 Conclusion

The Examination Entry System (EES) is a modernized solution aimed at streamlining university exam registration by automating applications, eligibility verification, and attendance sheet generation. By replacing manual processes with digital automation, it enhances efficiency, accuracy, and transparency.

While the system offers numerous benefits, addressing its limitations—such as internet dependency, system scalability, and security risks—will further strengthen its reliability.By implementing the recommended improvements, EES can become a more robust and user-friendly platform, ensuring smooth exam management for students, staff, and administrators.

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